Adic and Polyphase Quantum Refinement

Normalized packets, spectral elimination, and compatible limits

Arithmetic resolution and quantum refinement

The preceding constructions compare geometric and spectral resolutions. Arithmetic refinement provides another concrete family: finite unit registers, Haar packets, hierarchical Hamiltonians, and polyphase channels. The question remains the same. Which maps preserve the state, which preserve the dynamics, and what additional structure appears when hidden components are eliminated?

Normalized Haar averages identify coarse states inside a fine Hilbert space. Compatibility of the Hamiltonian, rather than normalization alone, determines whether that subspace carries the same evolution. Schur reduction and its time-domain memory equation describe the effect of components that cannot simply be discarded.

Infinite depth introduces a separate analytic problem. The hierarchical model specifies a common Hilbert space, a maximal operator domain, and the topology of approximation. Spectral decimation and hypergeometric companion functions provide exact coordinates within their declared domains. None of these identities alone identifies arithmetic depth with a physical observable.

The following account of covariant fiber dynamics supplies a candidate physical coupling and examines its energy and production constraints. That transition separates a mathematical refinement structure from a proposed model of matter.

The visual and verification appendices distinguish exact identities, finite computations, and formal coverage. The infinite arithmetic hierarchy is presented with its written proofs and is not assigned a Lean certificate beyond the statements specifically identified there.

Reading routes and conventions

The finite route begins with unit coordinates and Haar normalization. It continues through exact intertwining and controlled leakage. The spectral route begins with the block resolvent and follows radial chains, residue channels, and Chebyshev elimination. The limiting route begins with cylinder projections, constructs the maximal hierarchical Hamiltonian, and then treats compatible tensor dynamics.

An embedding and an elimination have different jobs. An isometry retains the old inner product. A unitary polyphase transformation retains the full fine operator in a new basis. A Schur complement solves for hidden coordinates on a regular spectral chart. A projection approximation converges only in the topology established by its proof. These operations should be identified before their consequences are compared.

Convention	Meaning
Hilbert normalization	Finite registers use an explicitly stated orthonormal basis. Cylinder functions use probability Haar measure. The conversion between counting coordinates and normalized indicators includes the square root of the cylinder measure.
Refinement	An isometry V has $V^*V = I$. Its old-range projector is $P = VV^*$ and its new-detail projector is $Q = I - P$. Exact dynamical compatibility is a separate identity, $H'V = VH$.
Hierarchy	The finite chapter introduces cumulative projectors E_r and their details. The infinite chapter writes E_m for conditional expectations and $\mathsf{F}_m = \mathsf{E}_m - \mathsf{E}_{m-1}$ for depth details, with the constant sector declared separately.
Self-adjointness	A finite Hamiltonian is Hermitian. Complex coupling amplitudes appear with conjugates in the reverse direction. An unbounded Hamiltonian also requires a dense domain and a proof that the declared realization is self-adjoint.
Spectral parameter	Schur formulas use $z - H$ and require the hidden resolvent $(z - QHQ)^{-1}$. Real-energy formulas are restricted to their regular chart. They are not evaluated at hidden poles without an extension argument.
Spectral coordinates	A residue or polyphase transformation is a change of basis. Selecting a channel, tracing a subsystem, averaging amplitudes, and eliminating hidden coordinates are different operations.
Limits	Strong vector, strong operator, graph-norm, and operator-norm convergence have distinct quantifiers. The infinite chapter proves the stated limits for its compatible block model.
Evidence	Ordinary derivations, fresh bounded calculations, retained program outputs, and archived formal source are recorded separately. A small residual does not prove a universal infinite-depth or physical claim.

The technical source concordance identifies statements by exact source member, formula, and claim. Each result here carries its own hypotheses. A source title or a recorded test result is not a substitute for those hypotheses.

Finite registers and normalized refinement

1 Finite unit registers and protected coordinates

Adic refinement first changes the address of a state. Its amplitudes remain complex numbers, and its measurement probabilities remain ordinary Born probabilities. For a finite address set X we use the counting inner product on $\ell^2(X;\mathbb{C})$, so the address vectors $|x\rangle$ are orthonormal. A bijection of address sets therefore gives a permutation unitary without changing the quantum probability rule. Arithmetic determines the address factorization. A Hamiltonian and a measurement prescription supply the additional physical choices.

The first coordinate issue is specific to units. For an odd prime p and an integer $h \geq 0$, set

$$U_{p,h} = (\mathbb{Z}/p^{h+1}\mathbb{Z})^{\times}.$$

For $p = 2$ the useful indexing starts one digit later:

$$U_{2,h} = (\mathbb{Z}/2^{h+2}\mathbb{Z})^{\times}.$$

The shifted dyadic index retains the sign even when the principal phase register has dimension one. It prevents the two units modulo four from being collapsed into one phase coordinate.

1.1 Torsion and principal units

Proposition 1.1 (Finite unit coordinates)**.** *For odd p, choose a unit $\zeta_{p,h}$ of order $p-1$ whose reduction generates $(\mathbb{Z}/p\mathbb{Z})^{\times}$. The maps*

$$(t,a) \longmapsto \zeta_{p,h}^{t}(1+p)^{a} \pmod{p^{h+1}}, \qquad (t,a) \in \mathbb{Z}/(p-1)\mathbb{Z} \times \mathbb{Z}/p^{h}\mathbb{Z}, \tag{1}$$

$$(s,a) \longmapsto (-1)^{s}5^{a} \pmod{2^{h+2}}, \qquad (s,a) \in \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2^{h}\mathbb{Z} \tag{2}$$

are group isomorphisms onto $U_{p,h}$ and $U_{2,h}$ respectively.

Proof. For odd p the valuation identity

$$v_p((1+p)^{p^k} - 1) = k + 1 \qquad (k \geq 0)$$

shows that $1 + p$ has order p^h modulo p^{h+1}. To check the identity, the case $k = 0$ is immediate. If $u = 1 + cp^{k+1}$ with $p \nmid c$, the binomial expansion of $u^p - 1$ has first term cp^{k+2}. Every remaining term has strictly larger p-valuation when p is odd. This proves the induction. The subgroup generated by $1 + p$ is contained in the kernel of reduction modulo p and has the same cardinality p^h, so it is that entire kernel.

For completeness, an order-$p-1$ lift of any nonzero residue can be chosen coherently. The root equation $x^{p-1} = 1$ has nonzero derivative $(p-1)x^{p-2}$ modulo p. If x_n is a root modulo p^n, substituting $x_n + cp^n$ modulo p^{n+1} determines one and only one residue c that cancels the error. Starting with a generator of $(\mathbb{Z}/p\mathbb{Z})^{\times}$ produces $\zeta_{p,h}$ with the required order. The cyclicity of this residue group follows from the elementary fact that a degree-d polynomial over a field has at most d roots: in a finite subgroup of the multiplicative group, the least common multiple m of the element orders is realized by an element. To see this, take for each prime divisor of m a component having the maximal prime-power order and multiply those commuting components of coprime orders. All elements solve $x^m = 1$.

Consequently the subgroup has size at most m, and Lagrange's theorem gives the reverse bound. The subgroup is cyclic.

If $\zeta_{p,h}^{t}(1+p)^{a}=1$, reduction modulo p gives $t=0$ in $\mathbb{Z}/(p-1)\mathbb{Z}$. The principal-unit order then gives $a=0$ in $\mathbb{Z}/p^{h}\mathbb{Z}$. Thus the displayed map is injective, and its domain has size $(p-1)p^{h}=|U_{p,h}|$, which proves surjectivity.

For the dyadic case, successive squaring gives

$$v_2(5^{2^k}-1)=k+2.$$

Indeed $5-1$ has valuation two, and for every such power the factor $5^{2^k}+1$ has valuation one. Thus 5 has order 2^h modulo 2^{h+2}. Its powers are exactly the 2^h units congruent to one modulo four. Multiplication by -1 supplies the units congruent to three modulo four. These two classes are disjoint, giving the claimed unique coordinates. □

In the odd-prime register t is a torsion coordinate and a is a principal phase coordinate. In the dyadic register s is a sign bit and a is again the principal phase. The Hilbert-space factorizations are therefore

$$\begin{aligned}\ell^2(U_{p,h}) &\cong \mathbb{C}^{p-1}_{\text{tors}}\otimes\ell^2(\mathbb{Z}/p^h\mathbb{Z}), && p \text{ odd},\\ \ell^2(U_{2,h}) &\cong \mathbb{C}^{2}_{\text{sign}}\otimes\ell^2(\mathbb{Z}/2^h\mathbb{Z}).\end{aligned} \tag{3}$$

These are unitary changes of basis. They do not assert that a generic state is a product state, or that a generic Hamiltonian is a sum of independent register Hamiltonians. Entanglement and interactions are preserved under the change of coordinates.

The lifting construction in the proof permits the coherent choice $\zeta_{p,h+1}\equiv\zeta_{p,h}\pmod{p^{h+1}}$. With that choice, reduction of unit addresses retains t and reduces a modulo p^h. If the torsion lifts are instead chosen independently at each depth, the finite factorizations remain exact, but a comparison between levels must include the resulting coordinate changes. A levelwise permutation is not by itself a specified refinement tower.

1.2 CRT readdressing and the Fourier dual

Let $\mathcal{P}_{\text{o}}$ be a finite set of odd primes and put

$$M=2^{h_2+2}\prod_{p\in\mathcal{P}_{\text{o}}}p^{h_p+1}.$$

The Chinese remainder theorem gives a bijection from the unit addresses modulo M to their local unit addresses. Combining it with Proposition 1.1 gives a permutation unitary

$$\mathcal{U}_{\text{CRT}}:\ell^2\big((\mathbb{Z}/M\mathbb{Z})^{\times}\big)\longrightarrow\mathbb{C}^{2}_{\text{sign}}\otimes\ell^2(\mathbb{Z}/2^{h_2}\mathbb{Z})\otimes\bigotimes_{p\in\mathcal{P}_{\text{o}}}\left(\mathbb{C}^{p-1}_{\text{tors}}\otimes\ell^2(\mathbb{Z}/p^{h_p}\mathbb{Z})\right). \tag{4}$$

For each basis vector, $\mathcal{U}_{\text{CRT}}|u\rangle$ is the basis vector with its unique tuple of coordinates. Distinct units have distinct tuples, so inner products are preserved. The dimensions also agree:

$$\varphi(M)=2^{h_2+1}\prod_{p\in\mathcal{P}_{\text{o}}}(p-1)p^{h_p}.$$

Multiplication of units becomes addition in each cyclic coordinate.

We use the positive-phase Fourier convention

$$(F_n)_{kx} = n^{-1/2} \exp(2\pi \mathrm{i} kx/n). \tag{5}$$

The character transform of the unit group consequently factors as

$$F_{U_M} = \mathcal{U}_{\mathrm{CRT}}^{*} \left[F_2 \otimes F_{2^{h_2}} \otimes \bigotimes_{p \in \mathcal{P}_{\mathrm{o}}} (F_{p-1} \otimes F_{p^{h_p}}) \right] \mathcal{U}_{\mathrm{CRT}}. \tag{6}$$

Here the unit-group character labels have been identified with the same cyclic coordinate sets. This is a transform on the multiplicative unit group, rather than the additive Fourier transform on all residues modulo M.

The additive transform has a related factorization, with different permutations on its input and its dual output. Let a, b be positive coprime integers and $N = ab$. Define

$$\begin{aligned} \mathcal{P}_{\mathrm{in}} |n\rangle &= |n \bmod a\rangle \otimes |n \bmod b\rangle, \\ \mathcal{P}_{\mathrm{out}} |k\rangle &= |kb^{-1} \bmod a\rangle \otimes |ka^{-1} \bmod b\rangle, \end{aligned}$$

where the inverses are taken in the indicated residue rings. Then

$$F_N = \mathcal{P}_{\mathrm{out}}^{*} (F_a \otimes F_b) \mathcal{P}_{\mathrm{in}}. \tag{7}$$

To verify the phase, let $n_a = n \bmod a$ and $n_b = n \bmod b$. At the dual coordinates $k_a = kb^{-1} \bmod a$ and $k_b = ka^{-1} \bmod b$, the product kernel is

$$N^{-1/2} \exp\left[2\pi \mathrm{i} \left(\frac{k_a n_a}{a} + \frac{k_b n_b}{b} \right) \right].$$

The numerator $bk_a + ak_b$ is congruent to k modulo both a and b, hence modulo ab. This kernel equals $N^{-1/2} \exp(2\pi \mathrm{i} kn/N)$, proving (7). Omitting the dual-output permutation would generally give a different phase labeling.

1.3 What is lost when the sign is removed

The linear address-forgetting map

$$C_{\mathrm{phase}} |s, a\rangle = |a\rangle \tag{8}$$

maps a 2^{h+1}-dimensional dyadic unit register onto a 2^h-dimensional principal phase register. Its kernel is the span of the 2^h vectors $|0, a\rangle - |1, a\rangle$. Thus

$$\operatorname{rank} C_{\mathrm{phase}} = 2^h, \qquad \dim \ker C_{\mathrm{phase}} = 2^h.$$

This is a rank-deficient linear sum of amplitudes. It is not an isometry, and it is not the quantum channel that discards a subsystem. For example it sends the normalized state $(|0, a\rangle + |1, a\rangle)/\sqrt{2}$ to $\sqrt{2}|a\rangle$. Its adjoint product therefore does not equal the identity.

Discarding the sign as a quantum operation instead takes the partial trace $\rho \mapsto \mathrm{Tr}_{\mathrm{sign}}\, \rho$. Both pure states $|0, a\rangle$ and $|1, a\rangle$ then produce the same reduced phase state $|a\rangle\langle a|$. Yet the sign observable

$$Z_{\mathrm{sign}} = (|0\rangle\langle 0| - |1\rangle\langle 1|) \otimes I$$

has expectations $+1$ and -1 on those states. No measurement on the retained phase state can reconstruct this difference. The obstruction is information loss in a stated map, not an alternative probability rule.

A separate arithmetic example shows why integral residue selectors and rational spectral idempotents should be distinguished. Let

$$A = \begin{pmatrix} 1 & 1 \\ 0 & -1 \end{pmatrix}, \qquad A^2 = I, \qquad E_{\pm} = \frac{I \pm A}{2}.$$

The identities $E_{\pm}^2 = E_{\pm}$, $E_+E_- = 0$, and $E_+ + E_- = I$ follow directly from $A^2 = I$. Both idempotents contain a denominator two, and the resultant of $x - 1$ and $x + 1$ has dyadic valuation one. Modulo two the two roots agree and $(A - I)^2 = 0$, while $A - I$ is nonzero. Thus the residual matrix can retain a nilpotent extension even though it no longer distinguishes the two rational eigenvalues. These $E_{\pm}$ are complementary algebraic spectral idempotents. Since A is not self-adjoint in the standard counting inner product, they are not orthogonal quantum measurement projectors in that inner product. The unit-address sign projector, by contrast, is an ordinary diagonal orthogonal projector.

2 Cylinder averages, Haar details, and digit analysis

The principal phase register has a canonical finite filtration. For a prime p and depth $h \geq 0$, set

$$G_{p,h} = \mathbb{Z}/p^h\mathbb{Z}, \qquad \mathcal{H}_{p,h} = \ell^2(G_{p,h}; \mathbb{C}).$$

At depth zero this is a one-dimensional space. A level-r cylinder contains all residues with a prescribed prefix modulo p^r, and has p^{h-r} members. Throughout this section depth refers to the number of retained least significant base-p digits.

2.1 Orthogonal averaging and its detail resolution

For $0 \leq r \leq h$, define the cylinder average

$$(E_r^{(h)} f)(x) = p^{-(h-r)} \sum_{y \equiv x \pmod{p^r}} f(y). \tag{9}$$

Its matrix is real and symmetric. Averaging twice over the same cylinder has the same effect as averaging once. Thus $E_r^{(h)}$ is the orthogonal projector onto functions constant on level-r cylinders. The range has dimension p^r.

Proposition 2.1 (Finite Haar resolution)**.** *The cylinder projectors satisfy*

$$E_r^{(h)} E_s^{(h)} = E_{\min(r,s)}^{(h)}, \qquad E_h^{(h)} = I.$$

Define

$$D_0^{(h)} = E_0^{(h)}, \qquad D_r^{(h)} = E_r^{(h)} - E_{r-1}^{(h)} \quad (1 \le r \le h).$$

Then the $D_r^{(h)}$ are mutually orthogonal projectors and

$$\begin{aligned} \sum_{r=0}^{h} D_r^{(h)} &= I, & D_r^{(h)} D_s^{(h)} &= \delta_{rs} D_r^{(h)}, \\ \operatorname{rank} D_0^{(h)} &= 1, & \operatorname{rank} D_r^{(h)} &= (p-1)p^{r-1} \quad (r \ge 1). \end{aligned} \tag{10}$$

Proof. If $r \le s$, the level-s cylinders partition each level-r cylinder into equally sized groups. Averaging over the smaller groups and then over the whole group returns the full average, giving $E_r E_s = E_r$. The self-adjointness of both sides also gives $E_s E_r = E_r$. This proves the minimum rule.

For $r \ge 1$, $(E_r - E_{r-1})^2 = E_r - 2E_{r-1} + E_{r-1} = D_r$. For $1 \le r < s$, expansion of $(E_r - E_{r-1})(E_s - E_{s-1})$ gives zero by the same minimum rule. The products with D_0 vanish as well. Telescoping gives the resolution of the identity. Finally $\operatorname{Ran} E_{r-1} \subset \operatorname{Ran} E_r$, so the detail rank is $p^r - p^{r-1}$, while E_0 has rank one. □

The decomposition is exact at each finite depth:

$$\mathcal{H}_{p,h} = \bigoplus_{r=0}^{h} \operatorname{Ran} D_r^{(h)}, \qquad \|f\|^2 = \sum_{r=0}^{h} \left\| D_r^{(h)} f \right\|^2. \tag{11}$$

A depth cutoff $m \le h$ retains $E_m^{(h)} = \sum_{r=0}^{m} D_r^{(h)}$. Its discarded norm is exactly

$$\left\| (I - E_m^{(h)}) f \right\|^2 = \sum_{r=m+1}^{h} \left\| D_r^{(h)} f \right\|^2.$$

No smoothness assumption is needed for this finite Parseval identity. A rate of decrease with m requires an additional bound on the detail weights of the state.

2.2 The child characters form the Haar basis

Write $\omega_p = \exp(2\pi \mathrm{i}/p)$. For $1 \le r \le h$, parent prefix $a \in \{0, \dots, p^{r-1}-1\}$, and character $q \in \{1, \dots, p-1\}$, define

$$\psi_{r,a,q}^{(h)}(x) = \begin{cases} p^{-(h-r+1)/2} \omega_p^{q d_r(x)}, & x \equiv a \pmod{p^{r-1}}, \\ 0, & \text{otherwise}, \end{cases} \tag{12}$$

where $d_r(x)$ is the digit multiplying p^{r-1}. Include the constant scaling vector $\psi_0^{(h)}(x) = p^{-h/2}$.

Each wavelet is constant on level-r cylinders and has zero average on its parent level-$r-1$ cylinder, because $\sum_{d=0}^{p-1} \omega_p^{qd} = 0$ for nonzero q. Thus it lies in $\operatorname{Ran} D_r^{(h)}$. Its support contains p^{h-r+1} points, giving

unit norm. Distinct parents have disjoint supports. For a fixed parent the inner product of the characters q, q' is

$$p^{-1} \sum_{d=0}^{p-1} \omega_p^{(q'-q)d} = \delta_{qq'}.$$

Wavelets at different levels are orthogonal because their detail projectors are orthogonal. There are exactly $(p-1)p^{r-1}$ wavelets at level r, equal to the rank in (10), so they span that detail subspace. Hence

$$D_r^{(h)} = \sum_{a=0}^{p^{r-1}-1} \sum_{q=1}^{p-1} |\psi_{r,a,q}^{(h)}\rangle\langle\psi_{r,a,q}^{(h)}|. \tag{13}$$

The omitted character $q = 0$ is the parent average. For $p = 2$ there is one average and one difference per parent. For odd p there is one average and $p - 1$ nontrivial child characters. The dyadic sign factor of (3) is independent of this average/detail split in the principal phase register.

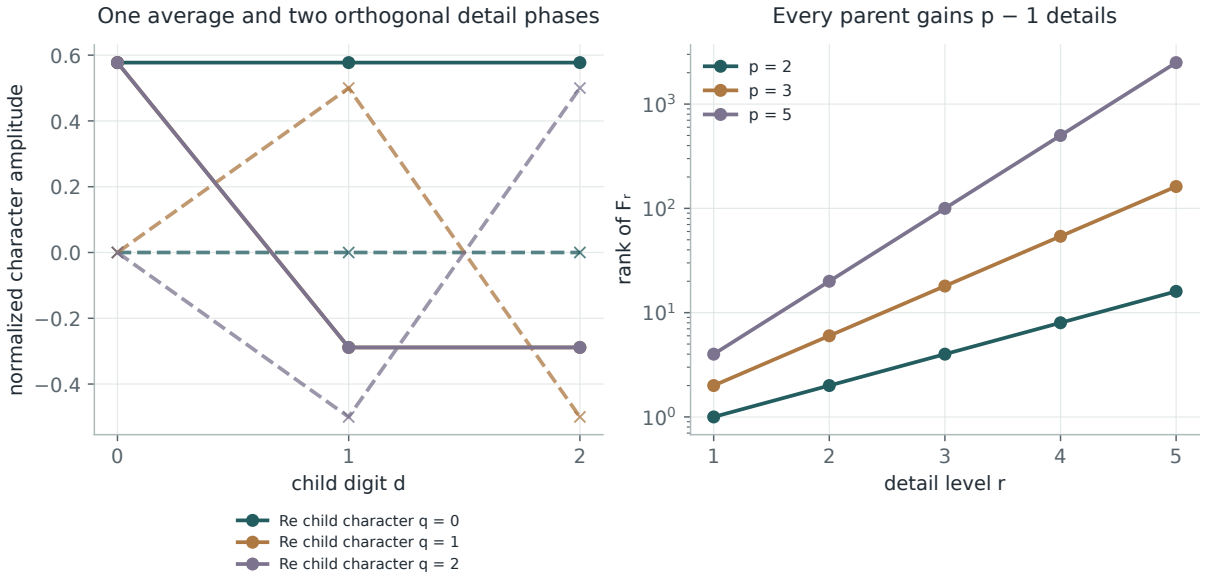


Figure 1. For $p = 3$, the normalized child characters contain one average and two orthogonal complex details. Solid curves show real parts and dashed curves show imaginary parts. The level rank $(p-1)p^{r-1}$ grows by the branching factor. The atlas uses F_r for the corresponding infinite cylinder detail; its finite restriction is $D_r^{(h)}$ for $r \le h$.

2.3 The refinement isometry and its adjoint

One more digit supplies p children for every address. The uniform refinement map is

$$(V_{p,h} f)(a + dp^h) = p^{-1/2} f(a), \quad 0 \le a < p^h, \quad 0 \le d < p. \tag{14}$$

In tensor coordinates it is $f \mapsto |+_p\rangle \otimes f$, where $|+_p\rangle = p^{-1/2} \sum_d |d\rangle$ and the child digit is listed first. The norm calculation is

$$\|V_{p,h} f\|^2 = \sum_a \sum_d p^{-1} |f(a)|^2 = \|f\|^2 .$$

The adjoint is the normalized amplitude sum

$$(V_{p,h}^* g)(a) = p^{-1/2} \sum_{d=0}^{p-1} g(a + dp^h). \tag{15}$$

Consequently

$$V_{p,h}^* V_{p,h} = I, \qquad V_{p,h} V_{p,h}^* = E_h^{(h+1)}, \qquad I - V_{p,h} V_{p,h}^* = D_{h+1}^{(h+1)}. \tag{16}$$

The first composition is an identity on the coarse space. The second is a projector on the fine space, not a fine-space identity.

Replicated vectors are constant in the new digit. All old cylinder averages commute with replication, giving

$$\begin{aligned} E_r^{(h+1)} V_{p,h} &= V_{p,h} E_r^{(h)} \quad (0 \le r \le h), \\ D_r^{(h+1)} V_{p,h} &= V_{p,h} D_r^{(h)} \quad (0 \le r \le h), \qquad\qquad D_{h+1}^{(h+1)} V_{p,h} = 0. \end{aligned} \tag{17}$$

Repeated refinements compose as isometries. For $w \ge 0$ and one coarse address,

$$V_{p,h+w-1} \cdots V_{p,h} |a\rangle = p^{-w/2} \sum_{\mathbf{d} \in \{0,\dots,p-1\}^w} |a, \mathbf{d}\rangle, \tag{18}$$

with the empty product interpreted as the identity. The factor $p^{-w/2}$ is forced by the isometry and orthonormality of the descendant addresses.

2.4 Digit reversal and the polyphase split

For $h \ge 1$, put $N = p^h$ and $m = p^{h-1}$. Two permutation unitaries separate different digits:

$$\begin{aligned} \mathcal{A}_{p,h}^{\mathrm{std}} |pk + r\rangle &= |r\rangle \otimes |k\rangle, & 0 \le r < p, \quad 0 \le k < m, \\ \mathcal{A}_{p,h}^{\mathrm{ad}} |a + dp^{h-1}\rangle &= |d\rangle \otimes |a\rangle, & 0 \le d < p, \quad 0 \le a < m. \end{aligned} \tag{19}$$

The standard polyphase split extracts the least significant residue. The adic split extracts the newest, most significant digit at the current depth. Let $\mathcal{R}_{p,h}$ reverse the h base-p digits, with $\mathcal{R}_{p,0} = I$ on the one-dimensional space. Then

$$\mathcal{A}_{p,h}^{\mathrm{ad}} = (I_p \otimes \mathcal{R}_{p,h-1}) \mathcal{A}_{p,h}^{\mathrm{std}} \mathcal{R}_{p,h}. \tag{20}$$

Indeed reversing $d_0 + d_1 p + \cdots + d_{h-1} p^{h-1}$ makes d_{h-1} the least significant digit. Standard analysis removes that digit, and reversing the remaining $h-1$ digits restores the prefix $a = d_0 + \cdots + d_{h-2} p^{h-2}$. This proves the identity on an orthonormal basis.

Let the new-digit modulation be $M_{p,h} |a + dp^{h-1}\rangle = \omega_p^d |a + dp^{h-1}\rangle$. The selector of child digit d is the exact roots-of-unity filter

$$R_d = \frac{1}{p} \sum_{\ell=0}^{p-1} \omega_p^{-d\ell} M_{p,h}^{\ell}. \tag{21}$$

On a child with digit e, the coefficient is $p^{-1} \sum_\ell \omega_p^{(e-d)\ell} = \delta_{de}$. Therefore $R_d R_e = \delta_{de} R_d$, $R_d^* = R_d$, and $\sum_d R_d = I$. These projectors select address digits. They are different from the projectors onto Fourier characters of a child register.

Apply the child Fourier transform after adic analysis:

$$\mathcal{W}_{p,h} = (F_p \otimes I_m)\mathcal{A}^{\mathrm{ad}}_{p,h}. \tag{22}$$

The zero child-frequency row of F_p is $p^{-1/2}(1, \dots, 1)$. Hence

$$\begin{aligned} (\langle 0| \otimes I_m)\mathcal{W}_{p,h} &= V^*_{p,h-1}, \\ \mathcal{W}^*_{p,h}[(I_p - |0\rangle\langle 0|) \otimes I_m]\mathcal{W}_{p,h} &= D^{(h)}_h. \end{aligned} \tag{23}$$

The second identity subtracts the coarse zero-frequency projector from the identity using (16). Recursive application to the zero-frequency channel alone constructs a unitary Haar analysis map $\mathcal{T}_{p,h}$ whose channel projectors Π_r have the ranks in (10) and obey

$$\mathcal{T}^*_{p,h}\Pi_r\mathcal{T}_{p,h} = D^{(h)}_r. \tag{24}$$

With our positive Fourier sign, the analysis row of character q pairs with the complex-conjugate wavelet character. This only permutes the nonzero character labels $q \leftrightarrow p - q$ and leaves the detail projectors unchanged.

The zero child-frequency channel retains a uniform superposition over the new digit. Retaining standard residue zero instead keeps one address out of each p addresses. Both spaces have the same dimension, but they are distinct subspaces. Their compressed dynamics and Schur complements consequently need not agree. Digit reversal and child Fourier analysis specify the transformations needed to compare them.

3 Which Hamiltonians are compatible with refinement?

An isometric address refinement does not automatically preserve a Hamiltonian. The exact condition is a block condition on that Hamiltonian. It can be checked without an approximation, a scaling limit, or a physical interpretation of the new digit.

Theorem 3.1 (Finite refinement classification)**.** *Let $V : \mathcal{H}_c \to \mathcal{H}_f$ be an isometry between finite Hilbert spaces. Set $P = VV^*$ and $Q = I - P$. For self-adjoint operators H_c, H_f, the following conditions are equivalent:*

$$H_f V = V H_c, \tag{25}$$

$$QH_fP = 0, \qquad PH_fP = VH_cV^*. \tag{26}$$

They hold exactly when, on $\operatorname{Ran} P \oplus \operatorname{Ran} Q$*,*

$$H_f = (VH_cV^*)|_{\operatorname{Ran} P} \;\oplus\; H_Q, \qquad H_Q = (QH_fQ)|_{\operatorname{Ran} Q}. \tag{27}$$

The detail operator H_Q may be any self-adjoint operator on $\operatorname{Ran} Q$*.*

Proof. Right multiplication of (25) by V^* gives $H_fP = VH_cV^*$. Left multiplication by Q and P gives (26). Conversely $PV = V$ and $V^*V = I$ imply

$$H_fV = (PH_fP + QH_fP)V = VH_cV^*V = VH_c.$$

Self-adjointness gives $PH_fQ = (QH_fP)^* = 0$. Thus both off-diagonal blocks vanish, and the remaining blocks are exactly those in (27). Conversely that direct sum has the two block properties. There is no further condition on the self-adjoint detail block. □

The distinction between invariance and compression is essential. Defining $H_c = V^*H_fV$ always matches the coarse block, but leaves QH_fV unconstrained. Thus compression alone does not establish (25). When it does hold, the coarse subspace is reducing, and fine evolution introduces no new detail amplitude from a coarse initial state.

Corollary 3.2 (Functional-calculus intertwining)**.** *Under Theorem 3.1, for every complex function f defined on the finite union of the spectra,*

$$f(H_f)V = Vf(H_c).$$

In particular, for real t, real β, and $z \notin \sigma(H_f)$,

$$\begin{aligned} e^{-\mathrm{i}tH_f}V &= Ve^{-\mathrm{i}tH_c}, \\ e^{-\beta H_f}V &= Ve^{-\beta H_c}, \\ (z - H_f)^{-1}V &= V(z - H_c)^{-1}. \end{aligned} \tag{28}$$

Proof. On Ran P, V is a unitary identification of H_c with the first block of H_f. The finite spectral decomposition therefore maps each coarse spectral subspace to the corresponding first-block spectral subspace. Applying f to the eigenvalues gives the identity. The three displayed functions are defined on the relevant spectra. □

The Euclidean operator $e^{-\beta H}$ in this corollary is a spectral multiplier. It is not generally unitary, and a normalized state after Euclidean evolution requires its own normalization. The identity does not compare normalized Gibbs states of the entire fine and coarse systems, since the fine partition function also counts the detail block. For a compatible finite tower, the coarse propagator and its embedded fine propagator agree exactly. This does not make an arbitrary interacting fine Hamiltonian compatible.

3.1 Hierarchical and translation-invariant examples

Choose a real sequence of level energies $\epsilon_{p,r}$ and define

$$H_{p,h}^{\mathrm{Haar}} = \sum_{r=0}^{h} \epsilon_{p,r} D_r^{(h)}. \tag{29}$$

Every wavelet in Ran $D_r^{(h)}$ has energy $\epsilon_{p,r}$. The projectors in (17) show directly that $H_{p,h+1}^{\mathrm{Haar}} V_{p,h} = V_{p,h} H_{p,h}^{\mathrm{Haar}}$. The new detail energy $\epsilon_{p,h+1}$ does not alter the embedded older levels. Moreover the recursive Haar transform diagonalizes the level blocks:

$$\mathcal{T}_{p,h} H_{p,h}^{\mathrm{Haar}} \mathcal{T}_{p,h}^* = \sum_{r=0}^{h} \epsilon_{p,r} \Pi_r.$$

This is exact for each finite h and every finite real choice of the energies. Their growth becomes a domain question only when an infinite-depth operator is constructed.

Let the cyclic translation be $(S_{p,h}f)(x) = f(x-1)$, with subtraction modulo p^h. Since a fine replicated function depends only on the residue modulo p^h,

$$S_{p,h+1}V_{p,h} = V_{p,h}S_{p,h}.$$

The adjoint translation obeys the same identity. Therefore the cyclic Laplacian $L_{p,h} = 2I - S_{p,h} - S_{p,h}^*$ intertwines, and for real κ_p the family

$$H_{p,h} = H_{p,h}^{\mathrm{Haar}} + \kappa_p L_{p,h} \tag{30}$$

is self-adjoint and refinement compatible. Translation permutes the level-r cylinders, so it commutes with each $E_r^{(h)}$ and $D_r^{(h)}$. The Laplacian thus preserves every detail subspace, although a chosen spatially localized Haar wavelet need not be its eigenvector.

This cyclic translation uses fixed unit displacement in the address group. Multiplying it by a depth-dependent factor intended as a physical inverse lattice spacing changes the tower. Such a factor is not covered by (30) unless the resulting operators still satisfy the stated intertwining equation. Exact adic compatibility and continuum spatial consistency are different requirements.

3.2 A valuation-shell propagator

For the pure hierarchical Hamiltonian, the finite spectral theorem gives

$$U_{p,h}(t) = e^{-\mathrm{i}tH_{p,h}^{\mathrm{Haar}}} = \sum_{r=0}^{h} e^{-\mathrm{i}t\epsilon_{p,r}} D_r^{(h)}. \tag{31}$$

The cylinder kernel is

$$E_r^{(h)}(x,y) = p^{-(h-r)}\mathbf{1}_{\{x\equiv y \,(\mathrm{mod}\, p^r)\}}.$$

Consequently

$$\begin{aligned} U_{p,h}(t;x,y) = {} & p^{-h}e^{-\mathrm{i}t\epsilon_{p,0}} \\ & + \sum_{r=1}^{h} e^{-\mathrm{i}t\epsilon_{p,r}} \left[p^{-(h-r)}\mathbf{1}_{x\equiv y \,(\mathrm{mod}\, p^r)} - p^{-(h-r+1)}\mathbf{1}_{x\equiv y \,(\mathrm{mod}\, p^{r-1})}\right]. \end{aligned} \tag{32}$$

Every indicator is determined by the truncated valuation $\min\{v_p(x-y), h\}$, with the diagonal assigned value h. The transition amplitude, and therefore the individual-address Born probability, is constant on each such valuation shell. At a fixed initial address y, the shell of valuation $k < h$ contains $(p-1)p^{h-k-1}$ final addresses. Its total probability includes this multiplicity. The diagonal has one address. Uniform probability per address on a shell does not mean equal total probabilities of different shells.

3.3 Leakage detects failure of compatibility

For a self-adjoint fine Hamiltonian with coarse projector P, set $B = QH_fP$. If a unit vector ψ lies in Ran P, its initial detail amplitude is

$$Qe^{-\mathrm{i}tH_f}\psi = -\mathrm{i}tB\psi + O(t^2), \qquad \left\|Qe^{-\mathrm{i}tH_f}\psi\right\|^2 = t^2\left\|B\psi\right\|^2 + O(|t|^3).$$

These are finite Taylor expansions for fixed H_f and ψ. Thus a nonzero B supplies a state with nonzero short-time leakage. The two-dimensional example

$$H_f = \begin{pmatrix} 0 & \overline{\gamma} \\ \gamma & 0 \end{pmatrix}, \qquad P = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$$

has compressed Hamiltonian zero but detail probability $\sin^2(|\gamma|t)$ from the first basis state. It fails exact refinement compatibility whenever $\gamma \neq 0$.

For z outside the spectrum of $H_Q = (QH_fQ)|_{\mathrm{Ran}\,Q}$, the coarse Schur self-energy is

$$\Sigma_P(z) = PH_fQ(z - H_Q)^{-1}QH_fP. \tag{33}$$

Compatibility makes it identically zero by setting $B = 0$. This zero is an algebraic statement, not a uniform bound on a resolvent near its spectrum. A nonzero coupling requires the energy-dependent Schur term when the detail sector is eliminated. The spectral part of this monograph develops that elimination and its time-domain memory.

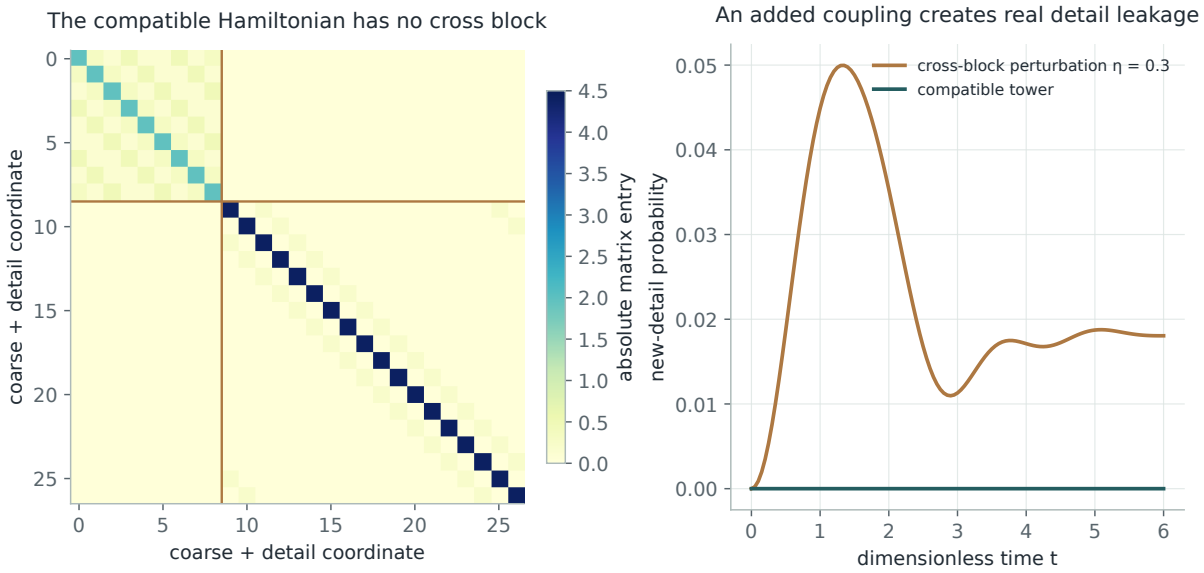


Figure 2. A finite compatible Hamiltonian has zero blocks between its embedded coarse space and its new detail space. In the plotted example, adding an off-diagonal coupling creates detail probability from a coarse initial state. This finite example illustrates the block criterion and supplies a negative control for unconditional refinement claims.

4 Normalization and collective coupling

Refinement fixes the norm of an embedded packet. It does not fix an interaction matrix element. The distinction becomes visible when one coarse packet couples to many descendants rather than to one already normalized aggregate packet. In this section $\hbar = 1$, so the Hamiltonian has units of inverse time.

4.1 Descendant weights are fixed by the isometry

After w refinements a packet has $n = p^w$ orthonormal descendants. Its normalized radial vector is

$$|R_{p,w}\rangle = n^{-1/2} \sum_{\mathbf{d}} |\mathbf{d}\rangle. \tag{34}$$

The norm is $n^{-1} \sum_{\mathbf{d}} 1 = 1$, so each descendant has amplitude modulus $p^{-w/2}$ and Born probability p^{-w}. This reproduces (18). The words "radial" and "angular" here refer to the uniform vector and its orthogonal complement in a finite descendant space. They do not introduce a spacetime radius or a rotation group.

Introduce a visible address $|c\rangle$ orthogonal to all descendants. Suppose every descendant has the same coupling from $|c\rangle$, with complex matrix element $\gamma_{p,w}$. The Hermitian interaction is

$$H_{\text{int}} = \sum_{\mathbf{d}} \left(\gamma_{p,w} |\mathbf{d}\rangle\langle c| + \overline{\gamma_{p,w}} |c\rangle\langle \mathbf{d}| \right). \tag{35}$$

The conjugate reverse coefficient is necessary when $\gamma_{p,w}$ is not real. Define the collective coefficient

$$G_{p,w} = \sqrt{n}\, \gamma_{p,w}. \tag{36}$$

Then $H_{\text{int}}|c\rangle = G_{p,w}|R_{p,w}\rangle$ and $H_{\text{int}}|R_{p,w}\rangle = \overline{G_{p,w}}|c\rangle$. If $|A\rangle = \sum_{\mathbf{d}} a_{\mathbf{d}} |\mathbf{d}\rangle$ is orthogonal to $|R_{p,w}\rangle$, its coefficients satisfy $\sum_{\mathbf{d}} a_{\mathbf{d}} = 0$. Hence $H_{\text{int}}|A\rangle = 0$. Only the radial vector is bright under this equal-leaf interaction.

This statement concerns the interaction block. A separate descendant Hamiltonian can mix radial and angular vectors. To obtain an invariant two-dimensional radial subsystem, the full descendant Hamiltonian must preserve the radial line, and the full Hamiltonian must leave its angular complement uncoupled from $|c\rangle$. A degenerate hidden energy ϵI is one sufficient choice.

4.2 The exact bright two-level problem

Assume those invariance conditions and order the basis as $(|c\rangle, |R_{p,w}\rangle)$. With visible energy E_c and radial energy ϵ, the exact bright block is

$$H_{\text{bright}} = \begin{pmatrix} E_c & \overline{G}_{p,w} \\ G_{p,w} & \epsilon \end{pmatrix}. \tag{37}$$

Set $\overline{E} = (E_c + \epsilon)/2$, $\delta = (E_c - \epsilon)/2$, and $\Omega = (\delta^2 + |G_{p,w}|^2)^{1/2}$. Then $(H_{\text{bright}} - \overline{E} I)^2 = \Omega^2 I$, giving

$$e^{-\mathrm{i}tH_{\text{bright}}} = e^{-\mathrm{i}t\overline{E}} \left[I \cos(\Omega t) - \mathrm{i} \frac{H_{\text{bright}} - \overline{E} I}{\Omega} \sin(\Omega t) \right]. \tag{38}$$

At $\Omega = 0$ this formula is understood by continuous extension. Starting in $|c\rangle$, the radial probability is therefore

$$\Pr(R_{p,w}; t) = \frac{|G_{p,w}|^2}{\delta^2 + |G_{p,w}|^2} \sin^2\left(t \sqrt{\delta^2 + |G_{p,w}|^2} \right). \tag{39}$$

The probability is defined to be zero when $\Omega = 0$, equivalently when both δ and $G_{p,w}$ vanish. On resonance $E_c = \epsilon$ it reduces to $\sin^2(|G_{p,w}|t)$. The individual descendant probabilities are $\Pr(R_{p,w}; t)/p^w$ when the hidden amplitude stays radial.

Eliminating the hidden radial level gives

$$\Sigma_{p,w}(z) = \frac{|G_{p,w}|^2}{z - \epsilon}, \qquad z \neq \epsilon. \tag{40}$$

This is the exact finite Schur term. The full visible Green function, when its denominator is nonzero, is $[z - E_c - \Sigma_{p,w}(z)]^{-1}$. It is distinct from the self-energy itself, and the locations of its poles depend on both sectors.

4.3 The coupling exponent is a model choice

Let $g \in \mathbb{C}$ be fixed and choose $\gamma_{p,w} = gp^{-\alpha w}$ for a real exponent α. Then

$$G_{p,w} = gp^{(1/2-\alpha)w}, \qquad |G_{p,w}|^2 = |g|^2 p^{(1-2\alpha)w}. \tag{41}$$

The exact branching multiplicity produces the factor $p^{w/2}$. Thus the collective coefficient grows with w for $\alpha < 1/2$, is independent of w for $\alpha = 1/2$, and decays for $\alpha > 1/2$. The same cases govern the operator norm of this equal-leaf interaction: its two radial eigenvalues are $\pm|G_{p,w}|$, so $\|H_{\text{int}}\| = |G_{p,w}|$. They do not determine the boundedness of a different infinite operator with many interacting depth sectors.

Per-leaf choice	α	$\lvert G_{p,w}\rvert$	$\lvert G_{p,w}\rvert^2$
Constant matrix element	0	$\lvert g\rvert p^{w/2}$	$\lvert g\rvert^2 p^w$
Constant collective coupling	$1/2$	$\lvert g\rvert$	$\lvert g\rvert^2$
Uniform measure coefficient	1	$\lvert g\rvert p^{-w/2}$	$\lvert g\rvert^2 p^{-w}$

The measure coefficient $\alpha = 1$ assigns the number p^{-w} to each per-leaf matrix element after factoring out g. It gives a collective amplitude of size $p^{-w/2}$ and a self-energy numerator of size p^{-w}. The self-energy estimate is uniform only where $|z - \epsilon|$ stays bounded below. Likewise

$$\Pr(R_{p,w}; t) \leq |g|^2 t^2 p^{-w} \quad \text{when } \alpha = 1$$

follows from (39) and $|\sin u| \leq |u|$. It is a fixed-time bound. On resonance a time of order $p^{w/2}/|g|$ can still transfer order-one probability.

A model that directly couples $|c\rangle$ to one normalized hidden packet with matrix element gp^{-w} instead has a collective coefficient of that same size and a self-energy numerator $|g|^2 p^{-2w}$. It is a valid two-level model, but it is not the equal-leaf model (35). The number of descendant addresses alone does not select between these interaction geometries. A physical refinement law must specify the coupling, the retained hidden modes, their energies, and the observation scale.

5 Polyphase coordinates and folded bands

The finite Haar decomposition isolates an average and its new detail channels. A standard polyphase decomposition instead groups the entries of a cyclic lattice by their residue class. Both transformations preserve the full state space. An elimination begins only when some coordinates are solved for and removed. The spectral parameter then enters the remaining operator, and the eliminated degrees of freedom return as memory in time.

We use complex Hilbert spaces with inner product linear in the second argument and set $\hbar = 1$. All matrices in the finite sections act on their stated coordinate spaces. In particular, the inverse of a low channel block is a low channel operator, rather than a full space matrix.

5.1 Residue coordinates and exact block reconstruction

Fix an integer $p \geq 2$ and $M \geq 3$, and put $N = pM$. Primality is not needed for the identities in this section. For $0 \leq r < p$, define the isometry $J_r : \mathbb{C}^M \to \mathbb{C}^N$ by

$$J_r|k\rangle = |pk + r\rangle, \qquad 0 \leq k < M.$$

Then $J_r^* J_s = \delta_{rs} I_M$ and $\sum_r J_r J_r^* = I_N$. The standard analysis map $\mathcal{A}$ sends $|pk + r\rangle$ to $|r\rangle \otimes |k\rangle$. This is a permutation unitary. For any operator X on $\mathbb{C}^N$, its residue blocks $X_{rs} = J_r^* X J_s$ reconstruct it:

$$X = \sum_{r,s=0}^{p-1} J_r X_{rs} J_s^*, \qquad (\mathcal{A} X \mathcal{A}^*)_{rs} = X_{rs}. \tag{42}$$

The first equality follows by inserting $\sum_r J_r J_r^*$ on both sides of X. If $H = H^*$, its blocks satisfy $H_{rs}^* = H_{sr}$. If U is unitary, the same insertion gives

$$\begin{aligned} \sum_r U_{rs}^* U_{rt} &= J_s^* U^* U J_t = \delta_{st} I_M, \\ \sum_s U_{rs} U_{ts}^* &= J_r^* U U^* J_t = \delta_{rt} I_M. \end{aligned} \tag{43}$$

These are the block unitary identities. Every coordinate is retained.

For finite H, expansion of the matrix exponential proves

$$e^{-it\mathcal{A}H\mathcal{A}^*} = \mathcal{A} e^{-itH} \mathcal{A}^*.$$

Indeed $(\mathcal{A}H\mathcal{A}^*)^n = \mathcal{A}H^n\mathcal{A}^*$, and the series converges in matrix norm. The same transformation of each Kraus operator gives a unitarily equivalent quantum channel: if $\Phi(\rho) = \sum_\mu K_\mu \rho K_\mu^*$ with $\sum_\mu K_\mu^* K_\mu = I$, then

$$K_\mu^{\text{poly}} = \mathcal{A} K_\mu \mathcal{A}^*, \qquad \sum_\mu (K_\mu^{\text{poly}})^* K_\mu^{\text{poly}} = I.$$

This is a change of coordinates in a given channel. It does not supply a new dissipative dynamics.

Proposition 5.1 (Probability currents between residue channels). *Suppose* $i\dot{\psi}_r = \sum_s H_{rs}\psi_s$ *with* $H^*_{rs} = H_{sr}$. *The channel weights* $w_r = \|\psi_r\|^2$ *satisfy*

$$\dot{w}_r = \sum_s \mathcal{J}_{r\leftarrow s}, \qquad \mathcal{J}_{r\leftarrow s} = 2\,\mathrm{Im}\langle\psi_r, H_{rs}\psi_s\rangle, \qquad \mathcal{J}_{r\leftarrow s} = -\mathcal{J}_{s\leftarrow r}.$$

Consequently $\sum_r w_r$ *is constant.*

Proof. Differentiating the norm gives $\dot{w}_r = 2\,\mathrm{Re}\langle\psi_r, -i\sum_s H_{rs}\psi_s\rangle$, which is the stated sum of imaginary parts. Self-adjoint block symmetry implies $\langle\psi_s, H_{sr}\psi_r\rangle = \overline{\langle\psi_r, H_{rs}\psi_s\rangle}$. The currents therefore cancel in pairs, including a zero diagonal current. □

Here w_r is a probability or squared amplitude weight. Calling it an energy would require an additional energy observable and its balance law.

5.2 A Bloch symbol with a fixed shift convention

Let $S|n\rangle = |n+1 \pmod N\rangle$ and

$$H = \mu I + \kappa(S + S^*), \qquad \mu, \kappa \in \mathbb{R}. \tag{44}$$

Use the coarse Fourier matrix $(F_M)_{jk} = M^{-1/2}e^{-2\pi ijk/M}$ and $\theta_j = 2\pi j/M$. After $\mathcal{A}$, Fourier transformation in k, and the permutation from residue major to momentum major ordering, the operator becomes a direct sum of $p \times p$ symbols. To see their entries, note that S sends (r,k) to $(r+1,k)$ for $r < p-1$ and sends $(p-1,k)$ to $(0,k+1)$. The coarse shift has Fourier eigenvalue $e^{-i\theta_j}$. Thus define

$$P_p(\theta)e_r = \begin{cases} e_{r+1}, & r < p-1, \\ e^{-i\theta}e_0, & r = p-1. \end{cases} \qquad \mathsf{H}_p(\theta) = \mu I_p + \kappa(P_p(\theta) + P_p(\theta)^*). \tag{45}$$

The transformed H is $\bigoplus_{j=0}^{M-1}\mathsf{H}_p(\theta_j)$. In particular, $P_p(\theta)^p = e^{-i\theta}I_p$.

Proposition 5.2 (Exact folded cyclic spectrum). *For* $q = 0, \dots, p-1$, *the eigenvalues of* $\mathsf{H}_p(\theta)$ *are*

$$E_q(\theta) = \mu + 2\kappa\cos\left(\frac{\theta + 2\pi q}{p}\right). \tag{46}$$

At $\theta = \theta_j$, *the union of these* pM *eigenvalues, counting multiplicity, is exactly the spectrum of* (44).

Proof. Put $\alpha_q = (\theta + 2\pi q)/p$ and let the rth component of v_q be $p^{-1/2}e^{i\alpha_q r}$. On the interior components, $P_p(\theta)v_q = e^{-i\alpha_q}v_q$. The wrapped component has the same value because $e^{ip\alpha_q} = e^{i\theta}$. The v_q are orthonormal by the finite character sum. Applying $\mu I + \kappa(P_p + P_p^*)$ gives (46). Every fine Fourier index is uniquely $k = j + qM$. Its angle satisfies $2\pi k/(pM) = (\theta_j + 2\pi q)/p$. Substitution into the fine cyclic eigenvalue $\mu + 2\kappa\cos(2\pi k/N)$ proves the multiplicity statement. □

The same construction works for any circulant polynomial $H = \sum_\ell c_\ell S^\ell$, with symbol $\sum_\ell c_\ell P_p(\theta)^\ell$. It preserves all bands, whether or not their energies coincide. For $N = p^h$, digit reversal relates this standard residue split to the newest-digit adic split developed earlier. It is still necessary to distinguish the retained residue-zero sites from the retained Haar average channel. They are different subspaces.

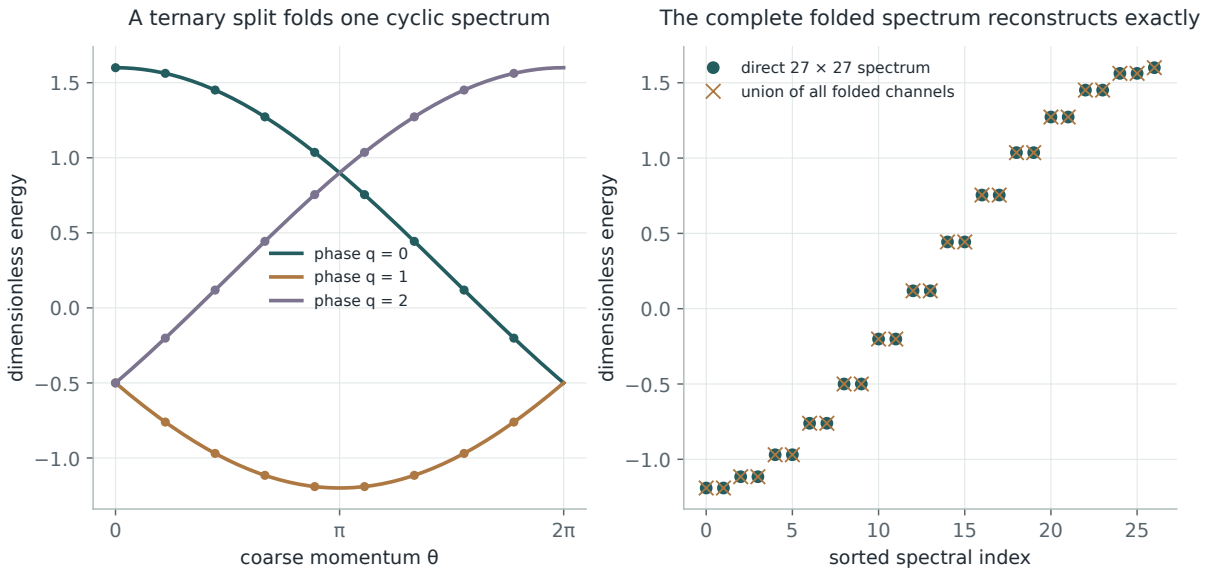


Figure 3. Three exact folded bands for the nearest-neighbor cycle. The discrete markers represent the finite cyclic spectrum. Band folding reorganizes its multiplicities without discarding any eigenstate.

6 Regular Schur reduction and causal memory

Let a finite Hilbert space be the orthogonal sum $\mathcal{H}_L \oplus \mathcal{H}_D$, and write a self-adjoint Hamiltonian as

$$H = \begin{pmatrix} A & B \\ B^* & D \end{pmatrix}, \qquad A = A^*, \quad D = D^*. \tag{47}$$

Here $B : \mathcal{H}_D \to \mathcal{H}_L$. Let ι_L embed $\mathcal{H}_L$ into the full space. This notation fixes the coordinate type of every compression.

6.1 The resolvent chart and its exceptional energies

Theorem 6.1 (Finite Feshbach identity). *If $z \notin \sigma(D)$, define*

$$\Sigma(z) = B(z - D)^{-1}B^*, \qquad \mathcal{S}(z) = z - A - \Sigma(z). \tag{48}$$

Then $z - H$ is invertible exactly when $\mathcal{S}(z)$ is invertible. On this common regular set,

$$\begin{aligned} \iota_L^*(z - H)^{-1}\iota_L &= \mathcal{S}(z)^{-1}, \\ \det(z - H) &= \det(z - D)\det \mathcal{S}(z). \end{aligned} \tag{49}$$

For $E \in \mathbb{R} \setminus \sigma(D)$, an eigenvector of H is equivalent to a nonzero solution of $\mathcal{S}(E)u = 0$, with hidden component $v = (E - D)^{-1}B^*u$.

Proof. Put $C = z - D$. Direct block multiplication gives

$$z - H = \begin{pmatrix} I & -BC^{-1} \\ 0 & I \end{pmatrix} \begin{pmatrix} \mathcal{S}(z) & 0 \\ 0 & C \end{pmatrix} \begin{pmatrix} I & 0 \\ -C^{-1}B^* & I \end{pmatrix}.$$

Both outer factors are invertible and have determinant one. This proves the invertibility equivalence and determinant formula. Inverting the factors gives the stated upper left block. Alternatively, in $(z - H)(u, v)^{\mathsf{T}} = (f, 0)^{\mathsf{T}}$, the lower equation gives $v = C^{-1}B^*u$ and the upper equation becomes $\mathcal{S}(z)u = f$. The same elimination with $f = 0$ proves the eigenvector statement. If $u = 0$ at such an E, invertibility of $E - D$ forces $v = 0$, so a nonzero full eigenvector indeed has $u \neq 0$. □

If $P_L = \iota_L \iota_L^*$, the full space compression is $P_L(z - H)^{-1}P_L = \iota_L \mathcal{S}(z)^{-1}\iota_L^*$. When a Haar transform $\mathcal{W}$ first exposes low and detail channels, the embedding in the original coordinates is $\iota_L = \mathcal{W}^*(|0\rangle \otimes I)$. The inverse on $\mathcal{H}_L$ must be embedded in this way before it is compared with a full space matrix.

The determinant equality is initially a rational identity on the regular set. Multiplication by $\det(z-D)$ cancels its denominators and yields the full characteristic polynomial, which extends across $\sigma(D)$. This extension does not make the displayed inverse of $z - D$ exist at a pole. At a resonant energy the uneliminated equations determine which hidden modes couple, survive, or cancel. A resonant subspace can be kept as part of an enlarged visible block.

6.2 Spectral residues, analytic sign, and branch weights

Let $D = \sum_\lambda \lambda E_\lambda$ be its spectral resolution. Then

$$\Sigma(z) = \sum_\lambda \frac{BE_\lambda B^*}{z - \lambda}. \tag{50}$$

Every residue $BE_\lambda B^*$ is positive semidefinite. Its rank records which directions of the hidden eigenspace reach the visible space. A vanishing residue removes the corresponding pole of Σ, even though $z - D$ is still singular there.

Proposition 6.2 (Upper half-plane sign and the exact visible norm)**.** *For $z = x + iy$ with $y > 0$,*

$$\operatorname{Im}\Sigma(z) = -yB\big((x - D)^2 + y^2 I\big)^{-1}B^* \leq 0, \qquad \operatorname{Im}\mathcal{S}(z) \geq yI.$$

In particular $\mathcal{S}(z)$ is invertible and $\left\|\mathcal{S}(z)^{-1}\right\| \leq y^{-1}$. For real $E \notin \sigma(D)$,

$$\Sigma'(E) = -B(E - D)^{-2}B^*.$$

If (u, v) is a normalized full eigenvector at E, then

$$1 = \langle u, (I - \Sigma'(E))u\rangle. \tag{51}$$

Proof. For real λ, $\operatorname{Im}(z - \lambda)^{-1} = -y/((x - \lambda)^2 + y^2)$. Applying this identity to the spectral sum gives the first formula and then the bound on $\operatorname{Im}\mathcal{S}$. For any u, Cauchy–Schwarz implies $y\|u\|^2 \leq |\langle u, \mathcal{S}(z)u\rangle| \leq \|u\|\,\|\mathcal{S}(z)u\|$. Thus $\mathcal{S}(z)$ has trivial kernel and an inverse of the stated norm in finite dimension. Differentiating the resolvent gives the minus sign in Σ'. The reconstructed hidden norm is $\|v\|^2 = \langle u, B(E - D)^{-2}B^*u\rangle$. Adding $\|u\|^2$ proves (51). □

For a one-dimensional visible block this says that an eigenstate's visible probability is $[1 - \Sigma'(E)]^{-1}$. It is at most one, as required. Eliminating a normalized eigenvector therefore changes its normalization metric as well as its spectral equation. Solving only a nonlinear energy equation does not provide that probability weight.

6.3 Exact elimination in the time domain

Theorem 6.3 (Causal memory with the initial hidden forcing)**.** *The finite system* $i\dot{u} = Au + Bv$, $i\dot{v} = Dv + B^*u$, *with initial data* (u_0, v_0), *is equivalent to*

$$\begin{aligned} i\dot{u}(t) &= Au(t) + Be^{-itD}v_0 - i\int_0^t K(t-s)u(s)\,ds, \\ K(\tau) &= Be^{-i\tau D}B^*, \qquad \tau \geq 0, \end{aligned} \tag{52}$$

together with reconstruction

$$v(t) = e^{-itD}v_0 - i\int_0^t e^{-i(t-s)D}B^*u(s)\,ds. \tag{53}$$

For $\operatorname{Im} z > 0$, *the self-energy is its exact one-sided transform:*

$$\Sigma(z) = -i\int_0^\infty e^{iz\tau}K(\tau)\,d\tau. \tag{54}$$

Proof. Differentiate $e^{itD}v(t)$ and integrate the hidden equation. This gives (53), whose substitution in the visible equation gives (52). Conversely the reconstruction differentiates to the hidden equation and supplies the prescribed initial value, so it recovers the full evolution. Since D is self-adjoint, $\|K(\tau)\| \leq \|B\|^2$. The factor $e^{-\operatorname{Im} z\,\tau}$ makes the transform absolutely convergent. On a spectral eigenvalue of D, $\int_0^\infty e^{i(z-\lambda)\tau}\,d\tau = i/(z-\lambda)$. The finite spectral sum then proves (54). □

The memory is causal because it uses only $s \leq t$. An initially empty hidden sector removes the forcing $Be^{-itD}v_0$, not the convolution. In finite dimension K is a finite sum of oscillatory terms. Its decay, a Markov approximation, or irreversible damping requires additional limits or assumptions. The closed full dynamics preserves $\|u(t)\|^2 + \|v(t)\|^2$, while the visible norm alone can decrease and later return. Energy-dependent Schur reduction is consequently different from replacing H by a fixed smaller self-adjoint matrix.

7 Rooted trees and radial Jacobi response

Let $T_{p,h}$ be a rooted p-ary tree of depth $h \geq 1$. Vertices at depth r have onsite energy $\epsilon_r \in \mathbb{R}$. Every parent-to-child edge from level r to $r+1$ has real coupling $\gamma_r/\sqrt{p}$, where $\gamma_r \in \mathbb{R}$. The Hamiltonian is the corresponding real symmetric adjacency matrix with onsite diagonal. The level symmetry and this branch normalization are structural assumptions of the model.

7.1 The radial reducing subspace

Define normalized level states

$$|r\rangle = p^{-r/2}\sum_{|v|=r}|v\rangle, \qquad r = 0, \dots, h.$$

They are orthonormal because different levels are disjoint and level r contains p^r vertices. A direct action calculation gives

$$H|r\rangle = \gamma_{r-1}|r-1\rangle + \epsilon_r|r\rangle + \gamma_r|r+1\rangle, \tag{55}$$

with omitted terms at the endpoints. For example, the coefficient at each child at level $r+1$ is $p^{-r/2}\gamma_r/\sqrt{p} = \gamma_r p^{-(r+1)/2}$. At each parent at level $r-1$, the p child contributions instead give $p\,p^{-r/2}\gamma_{r-1}/\sqrt{p} = \gamma_{r-1}p^{-(r-1)/2}$.

Let $R : \mathbb{C}^{h+1} \to \ell^2(T_{p,h})$ send e_r to $|r\rangle$. Equation (55) proves $HR = RJ_h$, where

$$J_h = \begin{pmatrix} \epsilon_0 & \gamma_0 & & & \\ \gamma_0 & \epsilon_1 & \gamma_1 & & \\ & \gamma_1 & \epsilon_2 & \ddots & \\ & & \ddots & \ddots & \gamma_{h-1} \\ & & & \gamma_{h-1} & \epsilon_h \end{pmatrix}. \tag{56}$$

Because H is self-adjoint, the orthogonal complement of $\operatorname{Ran} R$ is also invariant: for $w \perp \operatorname{Ran} R$, $\langle Re_r, Hw\rangle = \langle HRe_r, w\rangle = 0$. Hence this is a reducing decomposition, rather than only a compressed approximation. The root belongs to the radial block.

7.2 Angular chains and their multiplicities

There is a more precise form of the complementary decomposition. For each vertex u at depth $a < h$, choose an orthonormal basis of vectors $c = (c_0, \dots, c_{p-1}) \in \mathbb{C}^p$ with $\sum_d c_d = 0$. For $s = 0, \dots, h-a-1$, set

$$|u, c; s\rangle = p^{-s/2} \sum_{d=0}^{p-1} c_d \sum_{w \in \{0,\dots,p-1\}^s} |udw\rangle. \tag{57}$$

For fixed u, c, these are unit vectors on successive levels. They obey the Jacobi recurrence with onsite energies $\epsilon_{a+1}, \dots, \epsilon_h$ and couplings $\gamma_{a+1}, \dots, \gamma_{h-1}$. At $s = 0$ the upward amplitude into u is proportional to $\sum_d c_d$, so it vanishes. At $s > 0$ the same branch-counting calculation as in (55) produces the neighboring chain vectors.

Proposition 7.1 (Complete angular-chain decomposition)**.** *The radial chain and all the chains* (57) *form an orthogonal decomposition of $\ell^2(T_{p,h})$. A chain starting at depth r has length $h-r+1$ and appears with multiplicity $(p-1)p^{r-1}$, for $1 \le r \le h$.*

Proof. Within one chain, different s have disjoint level supports. Chains based at the same u and different contrast vectors have inner product $\langle c, c'\rangle = 0$ at a common level. If two vertices are in disjoint subtrees, their chains have disjoint support. If one vertex lies below another, the higher chain is constant, up to its fixed ancestral coefficient, on all children below the lower vertex at a common level. The lower chain has zero sum over its own child contrasts, so their inner product is zero. The same zero-sum calculation makes every angular vector orthogonal to the uniform radial level vector. At a fixed depth b, the radial vector contributes one dimension. The angular vectors present there contribute $\sum_{r=1}^{b}(p-1)p^{r-1} = p^b - 1$ dimensions. They are orthogonal and therefore form a basis of that p^b-dimensional level space. Taking all levels proves completeness. There are p^{r-1} choices of the vertex u at depth $r-1$ and $p-1$ contrasts at each, giving the stated multiplicity. □

The rank $(p-1)p^{r-1}$ thus counts chains born at level r or new Haar directions on a level. A single level's detail space is usually transported to the next level; it is the entire angular chain that is invariant under the tree Hamiltonian. The full dark dimension is

$$\sum_{r=1}^{h}(p-1)p^{r-1}(h-r+1)=\sum_{b=0}^{h}p^b-(h+1).$$

Every one of these states is invisible to a root-localized symmetric source because it is orthogonal to the radial reducing block.

7.3 Continued fractions and their meromorphic reading

For a tail $J_{r:h}$, put $g_r(z)=\langle r,(z-J_{r:h})^{-1}r\rangle$. On a common regular domain, scalar Schur elimination gives

$$\begin{aligned} g_h(z)&=(z-\epsilon_h)^{-1},\\ g_r(z)&=\frac{1}{z-\epsilon_r-\gamma_r^2 g_{r+1}(z)}, \qquad r<h. \end{aligned} \tag{58}$$

Defining $\Sigma_h=0$ and $\Sigma_r=\gamma_r^2/(z-\epsilon_{r+1}-\Sigma_{r+1})$ gives

$$\langle \text{root},(z-H)^{-1}\text{root}\rangle=g_0(z)=\frac{1}{z-\epsilon_0-\Sigma_0(z)}. \tag{59}$$

The equality to the tree resolvent follows from the exact reducing decomposition. Equations (58) are rational identities. Nested denominators can become singular at a real energy where a canceled final rational expression is finite. One should use the resolvent or the canceled rational function at such a point, rather than perform division by zero in the nested chart.

The Jacobi coefficients in (56) do not depend on p. Thus the root Green function and radial propagator are prime independent when the same ϵ_r,γ_r are used. The complete tree spectrum retains the p-dependent angular multiplicities. If the per-edge coupling is instead a fixed g_r, the radial coupling is $\sqrt{p}\,g_r$ and its response depends on p. Prime independence is therefore a consequence of the declared normalization, not a universality statement about arbitrary trees.

7.4 A visible two-component block

As a conditional finite model, take $A=H_D(k)=k\sigma_z+m_0\sigma_x$ with $k,m_0\in\mathbb{R}$ and $D=J_{1:h}\otimes I_2$. Couple the visible spinor identically to the first site of the tail by $B=\gamma_0(\langle 1|\otimes I_2)$. Its self-energy is scalar in spin space:

$$\Sigma_D(z)=\sigma(z)I_2, \qquad \sigma(z)=\gamma_0^2\langle 1,(z-J_{1:h})^{-1}1\rangle.$$

The determinant formula from Theorem 6.1 is

$$\det(z-H_{\text{full}})=\det(z-D)((z-\sigma(z))^2-k^2-m_0^2). \tag{60}$$

Consequently any eigenvalue in the regular chart obeys

$$(E-\sigma(E))^2=k^2+m_0^2. \tag{61}$$

The formula is a dressed dispersion relation of this specific model. Its mass, couplings, and hidden sector have been supplied. No physical interaction or measured dispersion follows solely from the tree split.

For a nondegenerate visible spin direction with normalized low-space direction u_*, the visible weight of its normalized full eigenstate is $[1-\sigma'(E)]^{-1}$. If $k = m_0 = 0$, the spin eigenspace is degenerate, but the same normalization relation holds for each chosen unit vector in that eigenspace.

If every internal tail hopping is nonzero, the first tail site is cyclic. One can see this without an abstract cyclicity theorem: a tail eigenvector with zero first component has zero second component by the first row, and successive rows force all later components to vanish. Therefore every tail eigenvector has nonzero first component. At an eigenvalue λ of that tail, solvability of $(\lambda - D)v = B^* u$ requires its right side to be orthogonal to the tail eigenspace. With $\gamma_0 \neq 0$ and identical spin coupling this forces $u = 0$. The visible equation would then require $Bv = 0$, which forces $v = 0$ on the tail eigenspace. Thus in this irreducible case the full coupled spectrum avoids the tail eigenvalues. With a vanishing hopping, coincident visible and hidden energies can occur and the regular-chart qualification in (61) is essential.

The scalar causal kernel is $K(\tau) = \gamma_0^2 \langle 1, e^{-i\tau J_{1:h}} 1\rangle$, multiplying I_2. It uses the same radial coefficients as the rational self-energy. Its exact causal relation is (54).

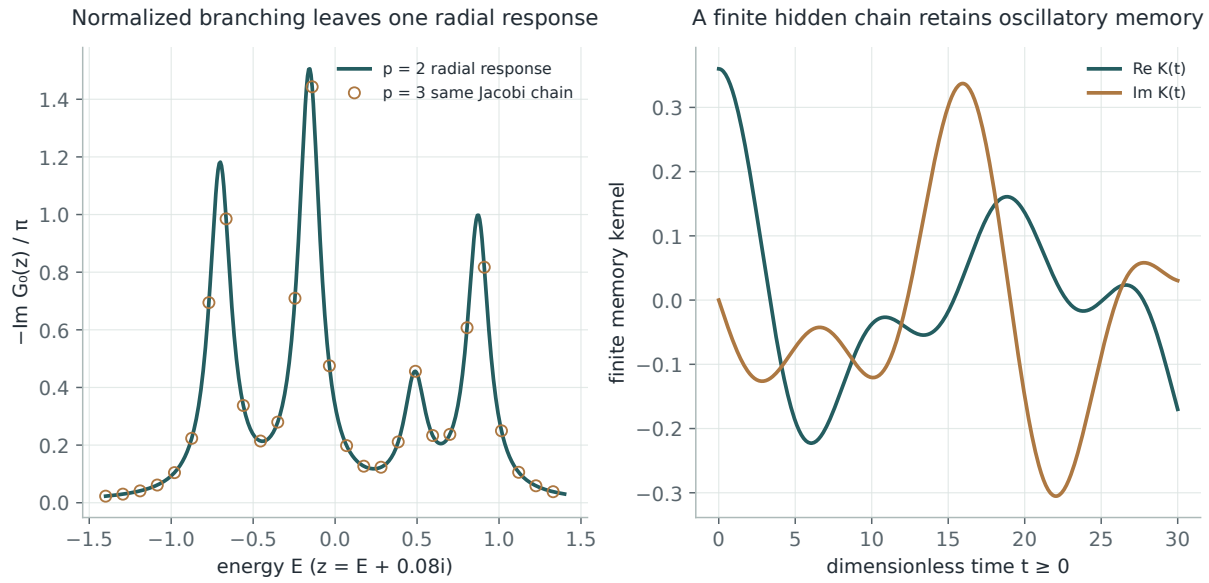


Figure 4. A branch-normalized finite tree has the same root response as its radial Jacobi chain. The complex-energy resolvent avoids real poles. The oscillatory memory is the return amplitude of a finite hidden chain.

8 Chebyshev decimation and resonant channels

We now eliminate intermediate residue sites in (44), assuming $\kappa \neq 0$. The retained sites are $C = \{0, p, \dots, (M-1)p\}$ and the eliminated sites comprise the $p-1$ interior sites between adjacent retained sites. Identify $\ell^2(C)$ with $\mathbb{C}^M$ and let $T|j\rangle = |j+1 \pmod M\rangle$. Set

$$x = \frac{z-\mu}{\kappa}, \qquad R_n(x) = 2T_n(x/2), \tag{62}$$

where T_n is the first-kind Chebyshev polynomial. The second-kind polynomials U_n satisfy $U_{-1} = 0$, $U_0 = 1$ and $U_{n+1}(t) = 2tU_n(t) - U_{n-1}(t)$.

8.1 The eliminated path fixes the Schur prefactor

The stationary equation on an interior site is $\psi_{n+1} = x\psi_n - \psi_{n-1}$. The eliminated block of $z - H$ in one cell is $\kappa K_{p-1}(x)$, where $K_n(x)$ has diagonal x and neighboring off-diagonal entries -1.

Lemma 8.1 (Dirichlet path determinant and endpoint entries). *For $n \geq 1$, $\det K_n(x) = U_n(x/2)$. If this determinant is nonzero,*

$$(K_n(x)^{-1})_{11} = (K_n(x)^{-1})_{nn} = \frac{U_{n-1}(x/2)}{U_n(x/2)}, \qquad (K_n(x)^{-1})_{1n} = \frac{1}{U_n(x/2)}.$$

For $n = 1$ these entries all denote the same value $1/x$.

Proof. Expansion along the last row gives the determinant recurrence $d_n = xd_{n-1} - d_{n-2}$, with $d_0 = 1$ and $d_1 = x$. This is exactly the recurrence and initial data for $U_n(x/2)$. Deleting the first or last row and column leaves K_{n-1}, proving the diagonal inverse entries by cofactors. For the endpoint entry, the cofactor consists of the product of $n-1$ entries -1 along the unique path from the first endpoint to the last. Its cofactor sign is $(-1)^{n+1}$, so the product has total sign $(-1)^{n+1}(-1)^{n-1} = 1$. Division by d_n gives the stated value. □

Theorem 8.2 (Exact free-cycle decimation). *For $U_{p-1}(x/2) \neq 0$, the Schur operator on the retained sublattice is*

$$\mathcal{S}_p(z) = \frac{\kappa}{U_{p-1}(x/2)}\big[R_p(x)I_M - (T + T^*)\big]. \tag{63}$$

Equivalently, $\mathcal{S}_p(z) = zI - H_{\text{eff}}^{(p)}(z)$, where

$$\begin{aligned} H_{\text{eff}}^{(p)}(z) &= \mu_p(z)I + \kappa_p(z)(T + T^*), \\ \kappa_p(z) = \frac{\kappa}{U_{p-1}(x/2)}, &\qquad \mu_p(z) = z - \frac{\kappa R_p(x)}{U_{p-1}(x/2)}. \end{aligned} \tag{64}$$

Proof. The eliminated sites split into M disjoint Dirichlet paths of length $p-1$. Hence their resolvent exists exactly when $U_{p-1}(x/2) \neq 0$. A retained site touches one endpoint on each of its two neighboring paths. Each path contributes the onsite self-energy $\kappa U_{p-2}/U_{p-1}$ by Lemma 8.1; the two contributions add. The path joining two adjacent retained sites contributes off-diagonal self-energy κ/U_{p-1}. Thus

$$\mathcal{S}_p(z) = \kappa\left(x - \frac{2U_{p-2}(x/2)}{U_{p-1}(x/2)}\right) I - \frac{\kappa}{U_{p-1}(x/2)}(T + T^*).$$

The identity $xU_{p-1}(x/2) - 2U_{p-2}(x/2) = 2T_p(x/2)$ follows from the three-term recurrence and $U_p - U_{p-2} = 2T_p$. For the latter identity, substituting $x/2 = \cos\theta$ gives $[\sin((p+1)\theta) - \sin((p-1)\theta)]/\sin\theta = 2\cos(p\theta)$, and polynomial equality extends it to all x. This proves (63) including its prefactor. Rearranging its diagonal gives (64). □

The normalized effective energy is $(z-\mu_p(z))/\kappa_p(z) = R_p(x)$. The parameter z remains fixed through the elimination. Since μ_p and κ_p depend on it, this is an energy-dependent spectral closure, not an energy-independent Hamiltonian flow. For $p = 2$, $R_2(x) = x^2 - 2$ and $U_1(x/2) = x$, giving

$$H^{(2)}_{\text{eff}}(z) = \mu I + \frac{\kappa^2}{z-\mu}(2I + T + T^*), \qquad z \neq \mu. \tag{65}$$

This decimation retains every other site. It does not retain the average of each neighboring pair.

8.2 The poles have an exact Dirichlet interpretation

The sine formula $U_{p-1}(\cos\theta) = \sin(p\theta)/\sin\theta$ shows that the excluded normalized energies are

$$x_j = 2\cos(j\pi/p), \qquad z_j = \mu + 2\kappa\cos(j\pi/p), \quad 1 \le j < p. \tag{66}$$

These are the energies of one eliminated path. Indeed its jth eigenvector has interior components $\sin(j\pi r/p)$, $r = 1, \dots, p-1$, and the nearest-neighbor recurrence gives precisely z_j. Both boundary values at $r = 0, p$ vanish. The eigenvectors are orthogonal and there are $p-1$ of them, so these exhaust the path spectrum.

The polynomial map R_p is defined at every x_j, while the Schur coefficients in (64) are not. Cancelling denominators in a determinant can cross this chart boundary; dividing by them in an eigenvector reconstruction cannot. A correct description there keeps the resonant cell modes or solves the full finite system. The fixed poles of the eliminated path are not all automatically eigenvalues of the full cyclic Hamiltonian.

Let Q project onto the eliminated sites. On the regular chart,

$$\det(z - H) = \det(z - QHQ)\det \mathcal{S}_p(z), \qquad \det(z - QHQ) = \left[\kappa^{p-1}U_{p-1}(x/2)\right]^M.$$

Their product is a polynomial identity after cancellation. It records both the hidden path factor and the coarse Schur factor, including their cancellations at the resonant energies.

8.3 Transfer coordinates and many-to-one branch maps

Define $A(x) = \begin{pmatrix} x & -1 \\ 1 & 0 \end{pmatrix}$. Its determinant is one and induction gives

$$A(x)^n = \begin{pmatrix} U_n(x/2) & -U_{n-1}(x/2) \\ U_{n-1}(x/2) & -U_{n-2}(x/2) \end{pmatrix}, \quad n \ge 1. \tag{67}$$

For $n = 1$ this uses $U_{-1} = 0$. Multiplication by $A(x)$ advances the first column through the Chebyshev recurrence and places the negative previous first column in the second. The trace is $R_n(x)$. Sampling the fine recurrence every p steps therefore uses the unimodular transfer matrix $A(x)^p$. Cayley–Hamilton gives the coarse recurrence $\chi_{j+1} - R_p(x)\chi_j + \chi_{j-1} = 0$ for $\chi_j = \psi_{pj}$. It explains the normalized spectral polynomial. The path-inverse proof above additionally supplies the prefactor needed for the resolvent.

The folded branches from Proposition 5.2, in normalized coordinates, satisfy

$$x_{p,q}(\theta) = 2\cos\left(\frac{\theta + 2\pi q}{p}\right), \qquad R_p(x_{p,q}(\theta)) = 2\cos\theta. \tag{68}$$

The second equality is just $T_p(\cos\alpha) = \cos(p\alpha)$. For a coarse normalized energy in the open interval $(-2, 2)$, choose its angle in $(0, \pi)$. These formulas supply the p distinct real roots of $R_p(x) = 2\cos\theta$. At the endpoints, some roots coincide and algebraic multiplicities must be retained. Globally the branches exchange labels when the angle crosses a cell boundary. There is no single-valued inverse of R_p on $[-2, 2]$.

For the fine cycle, $x_k = 2\cos(2\pi k/(pM))$ maps to $2\cos(2\pi k/M)$. Each coarse Fourier label has exactly p fine preimages. If two labels already have the same numerical energy, their multiplicities add as usual. Band folding retains the branch label and its vector, whereas normalized decimation identifies all of these labels at the same coarse energy. Their spectral relation does not make elimination an invertible operator change of basis.

9 Composition, transfer traces, and the free spectral measure

The spectral maps compose more simply than the energy-dependent operators that produce them. Keeping the normalized polynomial map, the rational Schur chart, and the transfer matrix distinct prevents an exact free-chain identity from being used outside its model.

9.1 Polynomial composition and iterated regular elimination

Theorem 9.1 (Chebyshev composition and Schur composition)**.** *For all positive integers m, n, the polynomial identities*

$$\begin{aligned} R_n(R_m(x)) &= R_{mn}(x) = R_m(R_n(x)), \\ U_{mn-1}(t) &= U_{n-1}(T_m(t))\, U_{m-1}(t) \end{aligned} \tag{69}$$

hold over $\mathbb{C}$. In a free cycle with $N = mnL$, $L \geq 3$, successive m- then n-site decimation at a fixed z agrees with direct mn-site decimation whenever its intermediate Schur inverses exist. Its final coefficients are

$$\kappa_{mn}(z) = \frac{\kappa}{U_{mn-1}(x/2)}, \qquad \mu_{mn}(z) = z - \frac{\kappa R_{mn}(x)}{U_{mn-1}(x/2)}. \tag{70}$$

Proof. For $x = 2\cos\theta$, first-kind composition gives $R_n(R_m(x)) = 2\cos(mn\theta) = R_{mn}(x)$. It holds on an interval and is therefore an identity of polynomials, including complex x. For $t = \cos\theta$ away from zeros of the sine denominators,

$$U_{n-1}(T_m(t))U_{m-1}(t) = \frac{\sin(mn\theta)}{\sin(m\theta)}\frac{\sin(m\theta)}{\sin\theta} = U_{mn-1}(t).$$

Polynomial equality extends it through those zeros and to all t. After the first regular elimination, the normalized variable is $x_m = (z - \mu_m)/\kappa_m = R_m(x)$. A second one gives

$$\kappa_{m,n} = \frac{\kappa_m}{U_{n-1}(x_m/2)} = \frac{\kappa}{U_{m-1}(x/2)U_{n-1}(T_m(x/2))} = \frac{\kappa}{U_{mn-1}(x/2)}.$$

Similarly $\mu_{m,n} = z - \kappa_{m,n}R_n(R_m(x))$, which gives (70). At each stage the matrices remain scalar nearest-neighbor circulants with the displayed complex coefficients, so their algebraic Schur formula continues to apply. These are rational identities on a common regular chart. They need not be interpreted as self-adjoint matrices at complex z. □

The product identity in (69) makes the exceptional denominators explicit. A nonzero final denominator implies both denominators along that two-stage chart are nonzero. At a zero of the final denominator the polynomial trace composition is still valid, but the divided Schur formulas require a new chart or retained resonant modes. Iterating one prime p through depth h, where the cycle has the required divisibility, replaces p by p^h in (70). Different prime orders give the same regular free-chain result. A nontranslation-invariant onsite potential, an interacting Hamiltonian, or a field-dependent constraint elimination needs its own closure analysis.

9.2 What remains true for a general unimodular transfer matrix

Let $M_0 \in SL_2(\mathbb{C})$ and $x = \operatorname{tr} M_0$. No diagonalizability assumption is needed. Cayley–Hamilton gives $M_0^2 - xM_0 + I = 0$, hence

$$M_0^n = U_{n-1}(x/2)M_0 - U_{n-2}(x/2)I, \qquad n \geq 1, \tag{71}$$

with $U_{-1} = 0$. For $n = 1$ it is the identity $M_0 = M_0$, and for $n = 2$ it is Cayley–Hamilton. Multiplying by M_0 and replacing M_0^2 by $xM_0 - I$ proves the induction step. Taking traces yields

$$\operatorname{tr}(M_0^n) = R_n(\operatorname{tr} M_0). \tag{72}$$

The coefficient identity is the same $xU_{n-1} - 2U_{n-2} = 2T_n$ used above.

If $w_{k+1} = M_0 w_k$, each residue subsequence satisfies

$$w_{nq+j} = M_0^j (M_0^n)^q w_0, \qquad 0 \leq j < n.$$

Thus ordinary matrix powering produces the normalized trace semigroup, including matrices with repeated eigenvalues at trace ± 2. The trace alone does not determine M_0, its eigenvectors, or a local Hamiltonian realization. In the free chain, $M_0 = A(x)$ has the required nearest-neighbor realization. For a general transfer problem only the matrix power and trace statements have been proved.

9.3 The arcsine measure and the Koopman operator

On $[-2, 2]$ define the probability measure

$$d\nu(x) = \frac{dx}{\pi\sqrt{4 - x^2}}. \tag{73}$$

It is the distribution of $2\cos\theta$ for a uniform angle on the circle, or on $[0, \pi]$. It is also the limiting normalized spectral measure of the free cycles: for a continuous test function f,

$$\frac{1}{N}\sum_{k=0}^{N-1} f(2\cos(2\pi k/N)) \longrightarrow \frac{1}{2\pi}\int_0^{2\pi} f(2\cos\theta)\, d\theta.$$

This is the Riemann-sum limit of a continuous periodic function.

Proposition 9.2 (Invariant free measure and exact mode routing). *For every integer $p \geq 2$, $(R_p)_*\nu = \nu$. The map $\mathcal{K}_p f = f \circ R_p$ is an isometry on $L^2(\nu)$. With*

$$\phi_0 = 1, \qquad \phi_n(x) = \sqrt{2}\, T_n(x/2), \quad n \geq 1,$$

its action and adjoint on an orthonormal basis are

$$\mathcal{K}_p \phi_n = \phi_{pn}, \qquad \mathcal{K}_p^* \phi_m = \begin{cases} \phi_{m/p}, & p \mid m, \\ 0, & p \nmid m. \end{cases} \tag{74}$$

In the first case $m = 0$ is included, so $\mathcal{K}_p^* \phi_0 = \phi_0$.

Proof. For a bounded measurable f, use the full circle parameterization:

$$\begin{aligned} \int f(R_p(x))\, d\nu(x) &= \frac{1}{2\pi} \int_0^{2\pi} f(2\cos(p\theta))\, d\theta \\ &= \frac{1}{2\pi p} \int_0^{2\pi p} f(2\cos u)\, du = \int f(x)\, d\nu(x). \end{aligned}$$

The last equality uses periodicity. Applying this equality to $|f|^2$ proves the isometry, first on bounded functions and then by truncation on all of $L^2(\nu)$. Pulling back to $[0, \pi]$ makes the ϕ_n the normalized cosine functions, and elementary cosine orthogonality proves their orthonormality. Their span is dense: Chebyshev polynomials span ordinary polynomials; those approximate continuous functions uniformly, for example by Bernstein polynomials after rescaling to $[0, 1]$. The approximation follows from uniform continuity and the binomial variance bound, by separating $|k/n - t| \leq \delta$ from its complement and then letting $n \to \infty$, $\delta \to 0$. Continuous functions are dense for this finite measure, as one sees by approximating interval steps in the angle coordinate by continuous ramps. This supplies completeness. Finally, Chebyshev composition gives $\mathcal{K}_p \phi_n = \phi_{pn}$. For every $n, m \geq 0$, $\langle \phi_n, \mathcal{K}_p^* \phi_m \rangle = \langle \phi_{pn}, \phi_m \rangle = \delta_{pn,m}$. Completeness yields the adjoint formula. □

$\mathcal{K}_p$ is an isometry with proper range, rather than a unitary: for example ϕ_1 is orthogonal to its range. Its adjoint discards nondivisible modes. Invariance of ν is a statement about the normalized free spectral distribution under R_p. It is not a proof that a sequence of growing Hilbert spaces has converged to an interacting Hamiltonian or that its quantum state converges. The direct-limit operator construction has separate hypotheses in the final part.

The mean logarithmic expansion of this map is also exact:

$$\int_{-2}^{2} \log |R_p'(x)|\, d\nu(x) = \log p. \tag{75}$$

Indeed for $x = 2\cos\theta$, $R_p'(x) = p \sin(p\theta)/\sin\theta$. Logarithms at its finitely many zeros are integrable, since they have the local form $\log|\theta - \theta_0|$ plus a bounded term. Hence the integral is $\log p$ plus the difference of the integrals of $\log|\sin(p\theta)|$ and $\log\sin\theta$ over $[0, \pi]$. The substitution $u = p\theta$ and π-periodicity of $\log|\sin u|$ make these two integrals equal. This calculation does not assert the entropy of a different physical dynamics.

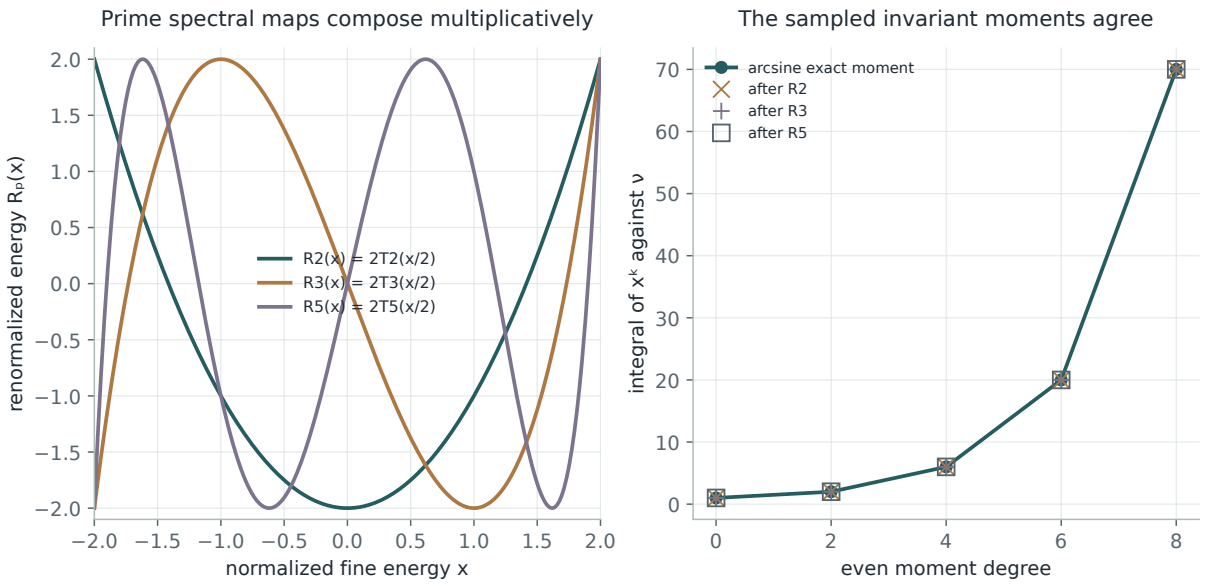


Figure 5. The normalized maps R_2, R_3, and R_5 preserve the same arcsine spectral measure. The accompanying bounded moment checks illustrate the exact measure identity, while each Schur realization retains its own excluded poles.

10 Analytic residue lanes and hypergeometric functions

Residue filtering also acts on coefficients of a convergent analytic series. Here the input is a function, not a quantum state, and the projector acts on its coefficient sequence. Evaluating one resulting lane at an operator does not generally turn that lane into a Hilbert space projector.

10.1 Character filtering and factorial separation

Fix an integer $L \geq 2$, write $\omega_L = e^{2\pi i/L}$, and let $F(z) = \sum_{n=0}^{\infty} a_n z^n$ be analytic for $|z| < \rho$. The whole disk is invariant under the rotations used below. Define

$$(\mathfrak{R}_{L,j}F)(z) = \frac{1}{L}\sum_{\ell=0}^{L-1} \omega_L^{-j\ell} F(\omega_L^{\ell} z), \quad 0 \leq j < L. \tag{76}$$

Absolute convergence for $|z| < \rho$ and the finite character sum give

$$(\mathfrak{R}_{L,j}F)(z) = \sum_{q=0}^{\infty} a_{Lq+j} z^{Lq+j}, \quad \mathfrak{R}_{L,j}\mathfrak{R}_{L,k} = \delta_{jk}\mathfrak{R}_{L,j}, \quad \sum_j \mathfrak{R}_{L,j} = I. \tag{77}$$

To prove the coefficient formula, the coefficient of $a_n z^n$ in (76) is $L^{-1}\sum_{\ell} \omega_L^{(n-j)\ell}$, equal to one exactly when $n \equiv j \pmod{L}$. The idempotent and reconstruction laws follow by filtering these coefficients again. For an entire F the formulas hold everywhere. For a function continued outside its disk, the domain and any chosen branches must also support all rotations; the local power-series identity does not remove such analytic restrictions.

Apply this filter to e^z and put

$$E_{L,j}(z) = \sum_{q=0}^{\infty} \frac{z^{Lq+j}}{(Lq+j)!}. \tag{78}$$

For $\alpha \in \mathbb{C}$, write $(\alpha)_q = \alpha(\alpha+1)\cdots(\alpha+q-1)$ and $(\alpha)_0 = 1$. Splitting the factorial into step-L factors gives

$$\begin{aligned} \frac{(Lq+j)!}{j!} &= \prod_{r=1}^{L} \prod_{k=0}^{q-1} (j+r+Lk) \\ &= L^{Lq} \prod_{r=1}^{L} \left(\frac{j+r}{L}\right)_q . \end{aligned} \tag{79}$$

Exactly one bottom parameter is one, at $r = L-j$. Define the multiset

$$\mathcal{A}_{L,j} = \left\{ \frac{j+r}{L} : 1 \le r \le L \right\} \setminus \{1\}.$$

Every member is positive, so none is a forbidden nonpositive integer in a hypergeometric denominator. Since $(1)_q = q!$,

$$E_{L,j}(z) = \frac{z^j}{j!}\, {}_0F_{L-1}\left(\begin{matrix} - \\ \mathcal{A}_{L,j} \end{matrix} ; \frac{z^L}{L^L} \right). \tag{80}$$

Here ${}_0F_s(;b_1,\dots,b_s;w)$ is defined by $\sum_{q\ge 0} w^q/[q!\prod_r (b_r)_q]$. The ratio of successive terms has magnitude $|w|/[(q+1)\prod_r |q+b_r|]$, which tends to zero. Its series is entire in w, so (80) is an everywhere convergent identity with the factorial series. For example $\mathcal{A}_{3,1} = \{2/3, 4/3\}$ and $E_{3,1}(z) = z\,{}_0F_2(;2/3,4/3;z^3/27)$.

10.2 The cyclic companion system

Termwise differentiation on compact subsets gives $E'_{L,j} = E_{L,j-1}$ for $j \ge 1$. For $j = 0$, the constant term vanishes under differentiation and reindexing gives $E'_{L,0} = E_{L,L-1}$. Let $\mathbf{E}_L = (E_{L,0},\dots,E_{L,L-1})^{\mathsf{T}}$ and define the cyclic matrix by $(C_L v)_0 = v_{L-1}$ and $(C_L v)_j = v_{j-1}$ for $j \ge 1$. Then

$$\mathbf{E}'_L = C_L \mathbf{E}_L, \qquad \mathbf{E}_L(0) = e_0, \qquad C_L^L = I, \qquad \mathbf{E}_L(z) = e^{zC_L} e_0. \tag{81}$$

One can verify the last equality directly: $C_L^n e_0 = e_{n \bmod L}$, so its jth exponential coordinate is exactly (78). Character filtering also gives

$$E_{L,j}(z) = \frac{1}{L} \sum_{\ell=0}^{L-1} \omega_L^{-j\ell} e^{\omega_L^{\ell} z}. \tag{82}$$

Thus factorial, hypergeometric, companion-state, and root-filter forms describe the same analytic function. Each form has a different numerical conditioning: a root sum can contain cancellations even when the defining series has positive terms.

For a general finite transfer matrix M_1, the residue-state identity $v_{Lq+j} = M_1^j (M_1^L)^q v_0$ follows by adding exponents. It needs no hypergeometric coefficients. A factorial lane provides a particular analytic coordinate system; it does not impose a unique microscopic Hamiltonian on every such transfer problem.

10.3 Bounded operators and controlled series errors

Theorem 10.1 (Entire lane calculus for a bounded operator)**.** *Let $A_0 \in \mathcal{B}(\mathcal{H})$ and $t \in \mathbb{C}$. The series $E_{L,j}(tA_0)$ converges absolutely in operator norm and satisfies*

$$\begin{aligned} E_{L,j}(tA_0) &= \frac{1}{L}\sum_{\ell=0}^{L-1} \omega_L^{-j\ell} e^{\omega_L^{\ell} tA_0}, \\ &= \frac{(tA_0)^j}{j!}\, {}_0F_{L-1}\left(\begin{matrix} - \\ \mathcal{A}_{L,j} \end{matrix}; \frac{(tA_0)^L}{L^L}\right), \\ \sum_{j=0}^{L-1} E_{L,j}(tA_0) &= e^{tA_0}. \end{aligned} \tag{83}$$

If all terms through degree Q are retained in a lane, the omitted operator norm is at most

$$e^a \frac{a^{Q+1}}{(Q+1)!}, \qquad a = |t|\,\|A_0\|\,. \tag{84}$$

Proof. The sum of norms of all exponential terms is at most e^a. A lane is a subseries of this sum, so it converges absolutely. The character calculation and factorial split can therefore be applied in operator norm, giving the first two identities. Partitioning the nonnegative degrees by their residues proves the third. All products are powers of the same operator, so there is no commuting-operator assumption. For the tail, bound a lane by the full tail. Since $(Q+1+k)! \geq (Q+1)!k!$,

$$\sum_{n=Q+1}^{\infty} \frac{a^n}{n!} \leq \frac{a^{Q+1}}{(Q+1)!} \sum_{k=0}^{\infty} \frac{a^k}{k!},$$

which proves (84). □

If $A_0^L = \alpha I$, the hypergeometric expression reduces every lane to a scalar function multiplying A_0^j. The propagator lies in the finite span of $I, A_0, \ldots, A_0^{L-1}$. A more general polynomial identity gives the corresponding finite algebra of powers. These statements follow from the supplied algebraic relation; they do not assert one for an arbitrary Hamiltonian.

The coefficient maps $\mathfrak{R}_{L,j}$ in (77) are idempotents on analytic functions. The evaluated operators $E_{L,j}(tA_0)$ generally are not. For $L = 2$, $E_{2,0}(tA_0) = \cosh(tA_0)$ and $E_{2,1}(tA_0) = \sinh(tA_0)$, which are neither an orthogonal projector family nor a pair of independent unitary evolutions. The sum reconstructs an evolution; an individual lane is an analytic contribution to it.

10.4 A squared-factorial lane for Bessel functions

The same separation gives the entire function

$$B(y) = \sum_{n=0}^{\infty} \frac{(-1)^n y^n}{(n!)^2} = {}_0F_1(;1;-y).$$

For $y \geq 0$, this is $J_0(2\sqrt{y})$; the entire series itself does not need a square-root branch. Define $B_{L,j}(y) = \sum_{q\geq 0}(-1)^{Lq+j}y^{Lq+j}/[(Lq+j)!]^2$. Squaring (79) gives

$$B_{L,j}(y) = \frac{(-1)^j y^j}{(j!)^2}\,{}_0F_{2L-1}\left(\begin{matrix} - \\ 1, \mathcal{A}_{L,j}, \mathcal{A}_{L,j}\end{matrix}; \frac{(-1)^L y^L}{L^{2L}}\right). \tag{85}$$

One of the two $q!$ factors is the conventional hypergeometric factorial and the other supplies the explicit bottom parameter 1. All bottom parameters are positive and the ratio test again gives entire convergence. Character filtering reconstructs $B = \sum_j B_{L,j}$. With the Euler derivative $\Theta = y\,d/dy$, termwise differentiation also gives the cyclic system

$$\Theta^2 B_{L,j}(y) = -yB_{L,j-1 \bmod L}(y). \tag{86}$$

For a term of positive degree n, its new coefficient is $n^2(-1)^n/(n!)^2 = -(-1)^{n-1}/[(n-1)!]^2$. The degree-zero term disappears under Θ^2, which also verifies the wraparound case $j = 0$. The scalar Gowdy Bessel mode therefore admits these analytic coordinates. Filtering that already-defined solution does not add a gravitational field equation.

11 Spectral domains and number-residue packets

The bounded calculus has two useful continuations. An unbounded Hamiltonian admits spectral functions with their exact domains. An oscillator admits actual residue projectors on its number spectrum. These are distinct constructions, and only the latter is a family of orthogonal quantum packet projectors.

11.1 An unbounded propagator does not make every lane bounded

Let H_∞ be self-adjoint with spectral measure E. For real t define the scalar functions $f_j(\lambda) = E_{L,j}(-it\lambda)$. The operator $f_j(H_\infty)$ has domain

$$\mathcal{D}(f_j(H_\infty)) = \left\{\psi : \int_{\mathbb{R}} |f_j(\lambda)|^2\, d\langle\psi, E(\lambda)\psi\rangle < \infty\right\}. \tag{87}$$

The scalar identity $\sum_j f_j(\lambda) = e^{-it\lambda}$ gives

$$\sum_j f_j(H_\infty)\psi = e^{-itH_\infty}\psi, \qquad \psi \in \bigcap_j \mathcal{D}(f_j(H_\infty)). \tag{88}$$

The equality holds on the stated common domain because it is a finite sum of spectral functions there. It does not say that each lane acts on every vector.

For an explicit example, take $L = 4$. The scalar formula gives

$$f_0(\lambda) = \tfrac{1}{2}(\cosh(t\lambda) + \cos(t\lambda)), \quad f_2(\lambda) = \tfrac{1}{2}(\cos(t\lambda) - \cosh(t\lambda)).$$

If H_∞ is multiplication by λ on $L^2(\mathbb{R})$ and $t \neq 0$, these functions define unbounded operators. Their sum cancels the growing hyperbolic contribution. The full propagator e^{-itH_∞} is nevertheless unitary on all of $L^2(\mathbb{R})$. For $L = 2$, by contrast, the two Schrödinger lanes are $\cos(tH_\infty)$ and $-i\sin(tH_\infty)$, both bounded. A larger lane count must not inherit this special dyadic boundedness without proof.

A sufficient common-domain condition is

$$\int_{\mathbb{R}} e^{2|t||\lambda|}\, d\langle\psi, E(\lambda)\psi\rangle < \infty, \tag{89}$$

since $|f_j(\lambda)| \le e^{|t||\lambda|}$. Spectral cutoffs $P_R = E([-R, R])$ supply a dense set satisfying this condition. On Ran P_R, the restricted Hamiltonian is bounded, the norm-series identities apply, and

$$\sum_j f_j(H_\infty) P_R \psi = e^{-itH_\infty} P_R \psi \longrightarrow e^{-itH_\infty}\psi.$$

On the common domain of the individual lanes, each term also converges to $f_j(H_\infty)\psi$ by its spectral-domain integrability. For an arbitrary vector only convergence of the summed cutoff expression is asserted. The domain question for an interacting Hamiltonian therefore remains even when a bounded surrogate has excellent numerical residuals.

11.2 A short factorial tail and its adic unit window

Let p be prime, let $L = mp$ for a positive integer m, and take $0 \le j < p$. For every $q \ge 0$,

$$g_q = \frac{j!(Lq)!}{(Lq+j)!} = \binom{Lq+j}{j}^{-1} \in \mathbb{Z}_p^{\times}. \tag{90}$$

Indeed $\binom{Lq+j}{j} = \prod_{r=1}^{j} (Lq + r)/r$. Each numerator and denominator is a p-adic unit because $Lq \equiv 0 \pmod p$ and $1 \le r < p$. For $j = 0$ the empty product is one. The proof also covers $p = 2$ in its short window. Outside this window, a factor can cross a multiple of p and the same proof gives no unit guarantee. This rational coefficient statement supplies no claim about the p-adic convergence of the complex exponential series.

Analytic refinement of a factorial lane is independent of this unit condition. Splitting $q = pr + a$, $0 \le a < p$, gives the exact identity

$$E_{L,j}(z) = \sum_{a=0}^{p-1} E_{pL,j+aL}(z), \qquad 0 \le j < L. \tag{91}$$

Absolute convergence permits this partition of the summation index. The labels $j + aL$ all lie in $[0, pL)$, but need not lie in the parent's short unit window. Exact complex analytic refinement and the local p-adic unit assertion therefore have separate hypotheses.

11.3 Actual orthogonal packets on the oscillator number spectrum

On the oscillator Fock space, let $\widehat{N}|n\rangle = n|n\rangle$ and define the bounded unitary $V = e^{2\pi i \widehat{N}/L}$. Its spectral residues are

$$Q_j^{(L)} = \frac{1}{L} \sum_{\ell=0}^{L-1} \omega_L^{-j\ell} V^\ell = \sum_{n \equiv j \pmod L} |n\rangle\langle n|. \tag{92}$$

On every number basis vector, the same finite character sum proves the second equality. These $Q_j^{(L)}$ are bounded orthogonal projectors, $Q_j Q_k = \delta_{jk} Q_j$, and $\sum_j Q_j = I$. No exponential norm series for the unbounded number operator is needed; its spectral calculus defines V.

For a coherent state

$$|\alpha\rangle = e^{-|\alpha|^2/2} \sum_{n=0}^{\infty} \frac{\alpha^n}{\sqrt{n!}} |n\rangle, \qquad \mu = |\alpha|^2 \geq 0,$$

the packet probability is the Poisson residue sum

$$\begin{aligned} p_{L,j}(\mu) &= \langle \alpha, Q_j^{(L)} \alpha \rangle = e^{-\mu} E_{L,j}(\mu) \\ &= \frac{1}{L} \sum_{\ell=0}^{L-1} \omega_L^{-j\ell} e^{\mu(\omega_L^\ell - 1)}. \end{aligned} \tag{93}$$

The factorial series proves nonnegativity, including the limiting values $p_{L,0}(0) = 1$ and $p_{L,j}(0) = 0$ for $j > 0$. Partition of the Poisson series proves $\sum_j p_{L,j} = 1$. Their equivalent hypergeometric form is $e^{-\mu}$ times (80) at $z = \mu$.

Proposition 11.1 (Approach to uniform residue probability)**.** *For fixed $L \geq 2$ and $\mu \geq 0$,*

$$\left| p_{L,j}(\mu) - \frac{1}{L} \right| \leq \frac{1}{L} \sum_{\ell=1}^{L-1} e^{-\mu(1-\cos(2\pi\ell/L))} \leq \frac{L-1}{L} e^{-\mu(1-\cos(2\pi/L))}. \tag{94}$$

Proof. In (93), the $\ell = 0$ term is $1/L$. The modulus of every other term is $e^{\mu(\operatorname{Re}\omega_L^\ell - 1)}/L$. The triangle inequality gives the first bound. Among the nonzero residues, the largest cosine is $\cos(2\pi/L)$, giving the second. □

The exponent is positive for each fixed L, but decreases as $2\pi^2/L^2$ when L grows. The conclusion is therefore uniform in the finite packet label j at fixed L; it is not a depth-uniform estimate at arbitrarily large residue modulus.

Finally the annihilation operator routes adjacent residues:

$$Q_r^{(L)} a Q_s^{(L)} = 0 \quad \text{unless } r \equiv s - 1 \pmod{L}, \qquad Q_r^{(L)} a^\dagger Q_s^{(L)} = 0 \quad \text{unless } r \equiv s + 1 \pmod{L}. \tag{95}$$

On a finite number-state combination this follows from $a|n\rangle = \sqrt{n}\,|n-1\rangle$ and $a^\dagger|n\rangle = \sqrt{n+1}\,|n+1\rangle$. Each residue projector preserves $\mathcal{D}(a)$ and $\mathcal{D}(a^\dagger)$, because it only removes terms from their weighted number norm. Truncation of a vector in either domain converges in the corresponding graph norm, so the identities extend to those domains. For the isolated oscillator Hamiltonian, which is a function of $\widehat{N}$, the packet projectors commute with the evolution and the weights stay constant. A driving term containing ladder operators can transfer weight according to the routing rule. Selecting such a drive is further model data.

These packet probabilities are directly interpretable within the ordinary oscillator model. A physical interpretation of adic digits, hidden tree levels, or a preferred hypergeometric residue split requires an independently specified coupling and observable. The exact analytic and spectral coordinates derived here remain useful without making that identification.

Compatible dynamics and infinite limits

12 One Hilbert space behind all cylinder depths

An isometric refinement map compares two finite registers. An infinite quantum dynamics requires more: a common completed Hilbert space, a specified operator domain, and a topology in which the cutoffs converge. These requirements can all be met for the cylinder tower. They are separate requirements for a walk on a growing finite tree.

Fix a prime p. The compact additive group $\mathbb{Z}_p$ has normalized Haar measure μ_p, with

$$B_{m,a} = a + p^m\mathbb{Z}_p, \qquad 0 \le a < p^m, \qquad \mu_p(B_{m,a}) = p^{-m}.$$

Write $\mathcal{H}_p = L^2(\mathbb{Z}_p, \mu_p; \mathbb{C})$. We use the complex inner product $\langle f, g\rangle = \int \overline{f} g \, d\mu_p$, linear in its second argument. The p-adic coordinate belongs to the measurable space. Amplitudes and the Born probabilities use ordinary complex Hilbert space.

12.1 Counting amplitudes and Haar densities

At depth m, the finite space $\mathcal{H}_{p,m} = \ell^2(\mathbb{Z}/p^m\mathbb{Z})$ has counting inner product and orthonormal basis e_a. Define

$$\mathcal{U}_{p,m} e_a = p^{m/2} \mathbf{1}_{B_{m,a}}, \qquad \mathcal{V}_m = \operatorname{span}\{\mathbf{1}_{B_{m,a}} : 0 \le a < p^m\}. \tag{96}$$

The cylinders are disjoint. Consequently

$$\langle \mathcal{U}_{p,m} e_a, \mathcal{U}_{p,m} e_b \rangle = p^m \mu_p(B_{m,a} \cap B_{m,b}) = \delta_{ab}.$$

The same indicators span $\mathcal{V}_m$, so $\mathcal{U}_{p,m}$ is unitary onto this subspace. For $c = \sum_a c_a e_a$, its image has constant value $p^{m/2} c_a$ on $B_{m,a}$. In particular,

$$\int_{B_{m,a}} |\mathcal{U}_{p,m} c|^2 \, d\mu_p = |c_a|^2. \tag{97}$$

There is no extra probability factor after this normalization.

The finite refinement map is

$$V_{p,m} e_a = \frac{1}{\sqrt{p}} \sum_{d=0}^{p-1} e_{a+dp^m}.$$

Since $B_{m,a} = \bigsqcup_d B_{m+1,a+dp^m}$, it satisfies

$$\mathcal{U}_{p,m+1} V_{p,m} = \mathcal{U}_{p,m}. \tag{98}$$

Indeed, the coefficient on each child indicator is $p^{-1/2} p^{(m+1)/2} = p^{m/2}$, precisely the coefficient on the parent. Thus the refinement changes the address resolution of an embedded state without changing the function in $\mathcal{H}_p$.

An equivalent finite convention uses the weighted norm

$$\|f\|^2_{m,\mathrm{Haar}} = p^{-m} \sum_a |f(a)|^2.$$

Here $f(a)$ is a density amplitude rather than a counting amplitude, and the embedding into $\mathcal{H}_p$ sends f to $\sum_a f(a)\mathbf{1}_{B_{m,a}}$. Its refinement repeats the parent value on all children, without $p^{-1/2}$. The coordinate change $f(a) = p^{m/2}c_a$ identifies the two conventions. The factors p^{-m} in the measure and $p^{m/2}$ in the counting embedding are a single normalization expressed in two places.

12.2 Cylinder averaging as an orthogonal projection

For $f \in \mathcal{H}_p$, define

$$(\mathsf{E}_m f)(x) = p^m \int_{x+p^m\mathbb{Z}_p} f(y)\, d\mu_p(y). \tag{99}$$

The integral is defined since L^2 is contained in L^1 on this probability space. It produces a function constant on each depth-m cylinder.

Proposition 12.1 (Nested cylinder projections)**.** *The operator E_m is the orthogonal projection onto $\mathcal{V}_m$. For $m, n \geq 0$,*

$$\mathsf{E}_m^* = \mathsf{E}_m, \qquad \mathsf{E}_m^2 = \mathsf{E}_m, \qquad \mathsf{E}_m\mathsf{E}_n = \mathsf{E}_{\min(m,n)}. \tag{100}$$

Moreover $\mathsf{E}_m f \to f$ in L^2 for every $f \in \mathcal{H}_p$.

Proof. For a cylinder-constant g, the integral of $f - \mathsf{E}_m f$ over each depth-m cylinder is zero. Multiplying by the constant value of $\overline{g}$ there and summing shows $\langle g, f - \mathsf{E}_m f\rangle = 0$. The image is in $\mathcal{V}_m$, and averaging a function in $\mathcal{V}_m$ returns the function. These statements characterize the orthogonal projection and establish the first two identities. The spaces are nested, because every cylinder partitions into its children. Orthogonal projections onto nested subspaces satisfy the third identity: the larger projection fixes the smaller range, and the reverse order follows by taking adjoints.

For completeness, the union of these ranges is dense. Cylinders form a countable clopen base for the topology on $\mathbb{Z}_p$ and generate its Borel sigma-algebra. If $f \in L^2$ is orthogonal to every cylinder indicator, the finite complex measure $\nu(A) = \int_A f\, d\mu_p$ vanishes on cylinders and their finite unions. The class of sets on which ν is zero contains the whole space, is closed under complements, and is closed under countable disjoint unions. The elementary monotone-class extension from the cylinder algebra therefore gives $\nu(A) = 0$ for every Borel set. Applying this to the positive and negative level sets of the real and imaginary parts of f yields $f = 0$ almost everywhere. The orthogonal complement of $\bigcup_m \mathcal{V}_m$ is thus zero.

Given $\varepsilon > 0$, choose $g \in \mathcal{V}_M$ with $\|f - g\|_2 < \varepsilon$. For $m \geq M$, the projection fixes g, so $\|f - \mathsf{E}_m f\|_2 = \|(I - \mathsf{E}_m)(f - g)\|_2 \leq \|f - g\|_2 < \varepsilon$. This proves strong convergence without exchanging an infinite integral with a limit. □

The completion of the isometric direct system $(\mathcal{H}_{p,m}, V_{p,m})$ is therefore $\mathcal{H}_p$: equation (98) gives a single isometric map on the algebraic direct limit, and its image is the dense union of the $\mathcal{V}_m$. For $m \leq M$, with $V_{p,m\to M} = V_{p,M-1}\cdots V_{p,m}$, one also has

$$\mathsf{E}_m\big|_{\mathcal{V}_M} = \mathcal{U}_{p,M} V_{p,m\to M} V_{p,m\to M}^* \mathcal{U}_{p,M}^*. \tag{101}$$

Both sides are the orthogonal projection in $\mathcal{V}_M$ onto $\mathcal{V}_m$. This is the exact relation between the finite matrices and the common infinite Hilbert space.

12.3 Orthogonal details and a complete Haar basis

Use $\mathsf{F}_0 = \mathsf{E}_0$ and $\mathsf{F}_m = \mathsf{E}_m - \mathsf{E}_{m-1}$ for $m \geq 1$. These are the infinite depth details called D_m in the archived chapter. They correspond to the finite $D_m^{(M)}$ after the embedding $\mathcal{U}_{p,M}$. Equation (100) gives

$$\mathsf{F}_m \mathsf{F}_n = \delta_{mn} \mathsf{F}_m, \qquad \sum_{m=0}^{M} \mathsf{F}_m = \mathsf{E}_M. \tag{102}$$

For example, if $1 \leq m < n$, the four products in $(\mathsf{E}_m - \mathsf{E}_{m-1})(\mathsf{E}_n - \mathsf{E}_{n-1})$ are $\mathsf{E}_m, \mathsf{E}_m, \mathsf{E}_{m-1}, \mathsf{E}_{m-1}$ and cancel. For $m = n$ the same expansion is $\mathsf{E}_m - \mathsf{E}_{m-1}$. The cases involving $m = 0$ use $\mathsf{E}_0 \mathsf{E}_n = \mathsf{E}_0$.

Consequently

$$\mathcal{H}_p = \bigoplus_{m=0}^{\infty} \operatorname{Ran} \mathsf{F}_m, \qquad \|f\|_2^2 = \sum_{m=0}^{\infty} \|\mathsf{F}_m f\|_2^2 . \tag{103}$$

The sum of the projections is I in the strong operator topology. The detail ranks are $d_0 = 1$ and $d_m = (p-1)p^{m-1}$ for $m \geq 1$, the differences of the dimensions $p^m - p^{m-1}$.

For an explicit basis, put $\omega_p = \exp(2\pi i/p)$ and define

$$\psi_{m,a,q} = p^{(m-1)/2} \sum_{d=0}^{p-1} \omega_p^{qd} \mathbf{1}_{B_{m,a+dp^{m-1}}}, \qquad \begin{matrix} m \geq 1, \quad 0 \leq a < p^{m-1}, \\ 1 \leq q < p. \end{matrix} \tag{104}$$

Each has zero average over its parent, since $\sum_d \omega_p^{qd} = 0$, and lies in $\operatorname{Ran} \mathsf{F}_m$. Its norm squared is $p^{m-1} p\, p^{-m} = 1$. Distinct parents have disjoint support, while two characters at the same parent have inner product $p^{-1} \sum_d \omega_p^{(q'-q)d} = \delta_{qq'}$. The number of these orthonormal functions is exactly d_m, so they span that level. Together with $\psi_0 = 1$, they form a complete orthonormal basis by (103).

12.4 A statewise approximation rate

For $s > 0$, define the Haar regularity norm

$$\|f\|_{\mathfrak{H}_p^s}^2 = \sum_{m \geq 0} p^{2sm} \|\mathsf{F}_m f\|_2^2 , \qquad \mathfrak{H}_p^s = \{ f \in \mathcal{H}_p : \|f\|_{\mathfrak{H}_p^s} < \infty \}.$$

No identification with another Sobolev convention is required here. The definition itself gives the quantitative cutoff estimate.

Proposition 12.2 (Exact tail and regularity bound)**.** *For $f \in \mathcal{H}_p$ and every $M \geq 0$,*

$$\|(I - \mathsf{E}_M) f\|_2^2 = \sum_{m > M} \|\mathsf{F}_m f\|_2^2 . \tag{105}$$

If $f \in \mathfrak{H}_p^s$, then

$$\|(I - \mathsf{E}_M) f\|_2 \leq p^{-s(M+1)} \|f\|_{\mathfrak{H}_p^s} . \tag{106}$$

The convergence $\mathsf{E}_M \to I$ is strong and is not convergence in operator norm.

Proof. The omitted details are orthogonal, so Parseval gives the equality. For $m > M$, $p^{2sm} \geq p^{2s(M+1)}$. Multiplying the omitted summands by this lower bound proves the squared estimate, and taking square roots gives (106). At every finite M there is a unit vector in $\operatorname{Ran} \mathsf{F}_{M+1}$, on which $I - \mathsf{E}_M$ acts as the identity. Thus $\|I - \mathsf{E}_M\| = 1$ for every M. □

The rate is a bound on states in the displayed regularity class. For an arbitrary L^2 state the tail still tends to zero, but no common geometric rate follows. A finite array already in $\mathcal{V}_L$ has zero tail once $M \geq L$, which is a different statement from testing an infinite tail.

13 Self-adjoint dynamics and what a cutoff converges to

13.1 The maximal domain for level energies

Let $(\epsilon_m)_{m\geq 0}$ be any sequence of real numbers. Define

$$\mathcal{D}(H_\epsilon) = \left\{ f \in \mathcal{H}_p : \sum_{m\geq 0} |\epsilon_m|^2 \left\| \mathsf{F}_m f \right\|_2^2 < \infty \right\},$$
$$H_\epsilon f = \sum_{m\geq 0} \epsilon_m \mathsf{F}_m f. \tag{107}$$

The series converges in L^2 exactly on this domain. It is not necessary to bound the energies uniformly, but it is necessary to state the domain.

Theorem 13.1 (Real level energies define a self-adjoint operator)**.** *The operator in* (107) *is densely defined, closed, and self-adjoint. The finite cylinder functions form an operator core. It is bounded if and only if* $\sup_m |\epsilon_m| < \infty$*, and its spectrum is*

$$\sigma(H_\epsilon) = \overline{\{\epsilon_m : m \geq 0\}}. \tag{108}$$

Proof. The orthogonal decomposition (103) identifies f with the square-summable sequence $f_m = \mathsf{F}_m f$. Finite sequences belong to the domain, so it is dense. The real multipliers imply symmetry. To identify the adjoint, suppose $g \in \mathcal{D}(H_\epsilon^*)$ and write $k = H_\epsilon^* g$. Testing the adjoint identity against an arbitrary vector supported on level m gives $k_m = \epsilon_m g_m$. Since k is square summable, $\sum_m |\epsilon_m|^2 \left\| g_m \right\|^2 < \infty$. Thus $g \in \mathcal{D}(H_\epsilon)$ and the adjoint agrees with H_ϵ. The reverse inclusion is symmetry. This proves self-adjointness and therefore closedness.

For $f \in \mathcal{D}(H_\epsilon)$, the cylinder truncation $\mathsf{E}_M f$ satisfies both $\left\| f - \mathsf{E}_M f \right\|_2 \to 0$ and $\left\| H_\epsilon (f - \mathsf{E}_M f) \right\|_2 \to 0$, because the two squared norms are the tails of convergent nonnegative series. This is precisely density in the graph norm and proves the core assertion. If the energies have supremum $C < \infty$, Parseval gives $\left\| H_\epsilon f \right\|_2 \leq C \left\| f \right\|_2$. Conversely, a unit vector on level m has image norm $|\epsilon_m|$, so boundedness forces a finite supremum.

If z is outside the closed energy set, the sequence $(\epsilon_m - z)^{-1}$ is bounded and defines an inverse for $H_\epsilon - z$. Its image lies in the domain because $\epsilon_m/(\epsilon_m - z) = 1 + z/(\epsilon_m - z)$ is bounded as well. This proves that such z is in the resolvent set. An energy ϵ_m is an eigenvalue, while a limit point z has unit level vectors u_j with $\left\| (H_\epsilon - z) u_j \right\|_2 \to 0$. A bounded inverse would contradict this limit. Hence the closed energy set is exactly the spectrum. □

This proof also excludes a hidden choice of boundary condition at infinite depth. On the finite-cylinder core, the real level operator has the unique self-adjoint closure just described.

13.2 Explicit spectral calculus and strong continuity

For bounded Borel $F : \mathbb{R} \to \mathbb{C}$, define

$$F(H_\epsilon)f = \sum_{m\geq 0} F(\epsilon_m)\mathsf{F}_m f. \tag{109}$$

The series converges by Parseval and $\|F(H_\epsilon)f\|_2 \leq \sup_m |F(\epsilon_m)|\,\|f\|_2$. The same level decomposition verifies sums, products, and adjoints of these multipliers directly. In particular,

$$U(t) = e^{-itH_\epsilon} = \sum_{m\geq 0} e^{-it\epsilon_m}\mathsf{F}_m, \tag{110}$$

$$(z - H_\epsilon)^{-1} = \sum_{m\geq 0}(z-\epsilon_m)^{-1}\mathsf{F}_m, \quad z \notin \sigma(H_\epsilon). \tag{111}$$

The projections in these sums converge strongly, not absolutely in operator norm in general.

The modulus-one multipliers give $U(t)^*U(t) = I$ and $U(t+s) = U(t)U(s)$. They also give strong continuity:

$$\|(U(t) - U(t_0))f\|_2^2 = \sum_m |e^{-it\epsilon_m} - e^{-it_0\epsilon_m}|^2\,\|\mathsf{F}_m f\|_2^2 \longrightarrow 0.$$

To justify this limit, first bound the omitted summands by $4\sum_{m>M}\|\mathsf{F}_m f\|_2^2$, then use ordinary continuity in the finite sum up to M. On $\mathcal{D}(H_\epsilon)$ the same argument, using $|(e^{-it\epsilon}-1)/t| \leq |\epsilon|$, proves

$$\lim_{t\to 0}\frac{U(t)f - f}{t} = -iH_\epsilon f.$$

Thus the declared self-adjoint operator is the generator. A general L^2 state evolves unitarily even if this derivative does not exist.

13.3 Two distinct extensions of the finite dynamics

Let $H_\epsilon^{(M)}$ be the finite restriction of H_ϵ to $\mathcal{V}_M$. One useful approximation to $U(t)$ is its embedded finite propagator, extended by zero on the omitted subspace:

$$U_M^{\mathrm{emb}}(t) = \sum_{m=0}^{M} e^{-it\epsilon_m}\mathsf{F}_m = U(t)\mathsf{E}_M. \tag{112}$$

It is unitary on $\mathcal{V}_M$ and has kernel $\mathcal{V}_M^\perp$ as an operator on $\mathcal{H}_p$.

Proposition 13.2 (Exact state error for a compatible cutoff)**.** *For every $f \in \mathcal{H}_p$,*

$$\left\|U(t)f - U_M^{\mathrm{emb}}(t)f\right\|_2 = \left\|(I - \mathsf{E}_M)f\right\|_2 . \tag{113}$$

This equality is uniform for all real t. For $f \in \mathfrak{H}_p^s$, the right side is bounded by $p^{-s(M+1)} \left\|f\right\|_{\mathfrak{H}_p^s}$. For $f \in \mathcal{D}(H_\epsilon)$ there is also the exact graph-norm identity

$$\begin{aligned} &\left\|U(t)f - U_M^{\mathrm{emb}}(t)f\right\|_2^2 + \left\|H_\epsilon(U(t)f - U_M^{\mathrm{emb}}(t)f)\right\|_2^2 \\ &\quad = \sum_{m>M} (1 + |\epsilon_m|^2) \left\|\mathsf{F}_m f\right\|_2^2 . \end{aligned} \tag{114}$$

Proof. The difference is the omitted orthogonal sum $\sum_{m>M} e^{-it\epsilon_m}\mathsf{F}_m f$. Its multipliers have modulus one, so Parseval gives (113). Applying H_ϵ multiplies the omitted level by ϵ_m and gives the second identity. The domain hypothesis ensures that this second series is finite. The regularity estimate is Proposition 12.2. □

An alternative is to extend the finite Hamiltonian by the zero Hamiltonian on the omitted subspace:

$$\widetilde{H}_M = H_\epsilon \mathsf{E}_M = \sum_{m \le M} \epsilon_m \mathsf{F}_m .$$

This is a bounded self-adjoint operator on all of $\mathcal{H}_p$. Its propagator is a full-space unitary, but has a different tail:

$$e^{-it\widetilde{H}_M} = U_M^{\mathrm{emb}}(t) + (I - \mathsf{E}_M), \qquad \left\|(e^{-it\widetilde{H}_M} - U(t))f\right\|_2 \le 2 \left\|(I - \mathsf{E}_M)f\right\|_2 . \tag{115}$$

Likewise, for nonreal z,

$$\begin{aligned} (z - \widetilde{H}_M)^{-1} &= \mathsf{E}_M (z - H_\epsilon)^{-1} + z^{-1}(I - \mathsf{E}_M), \\ \left\|((z - \widetilde{H}_M)^{-1} - (z - H_\epsilon)^{-1})f\right\|_2 &\le (|z|^{-1} + |\operatorname{Im} z|^{-1}) \left\|(I - \mathsf{E}_M)f\right\|_2 . \end{aligned} \tag{116}$$

Thus $\widetilde{H}_M$ converges to H_ϵ in the strong resolvent sense. These statements specify operators on one common Hilbert space. They do not compare matrices of changing dimensions without an embedding.

The embedded propagators converge strongly, uniformly in time on each fixed state. Their operator norm error nevertheless satisfies

$$\left\|U(t) - U_M^{\mathrm{emb}}(t)\right\| = 1 \qquad \text{for every } M, t.$$

The fully unitary extensions also converge strongly with the bound in (115); norm convergence of these extensions needs a separate condition on the omitted phases. Neither assertion is a general norm-limit theorem for arbitrary quantum dynamics.

13.4 A graph-domain example with exact infinite tails

Example 13.3 (Convergent states and finite energy derivatives)**.** Choose one unit vector $u_m \in \operatorname{Ran} \mathsf{F}_m$ at every level, let $0 < q < 1$, and set

$$f_q = \sum_{m \ge 0} \sqrt{(1-q)q^m}\, u_m, \qquad \epsilon_m = 2^m .$$

Then $\|f_q\|_2 = 1$ and

$$\|(I - \mathsf{E}_M) f_q\|_2^2 = q^{M+1}. \tag{117}$$

However, $f_q \in \mathcal{D}(H_\epsilon)$ exactly when $4q < 1$:

$$\|H_\epsilon f_q\|_2^2 = (1-q) \sum_{m \geq 0} (4q)^m.$$

For $q < 1/4$, its graph tail is exactly

$$q^{M+1} + \frac{(1-q)(4q)^{M+1}}{1-4q}. \tag{118}$$

For $q = 1/8$, the state norm tail decays faster than the graph norm tail. For $q = 1/2$, the normalized state still has a convergent L^2 cylinder approximation and a well-defined unitary evolution, but is outside the Hamiltonian domain. All these conclusions concern genuinely infinite series, rather than a finite last level with its tail set to zero.

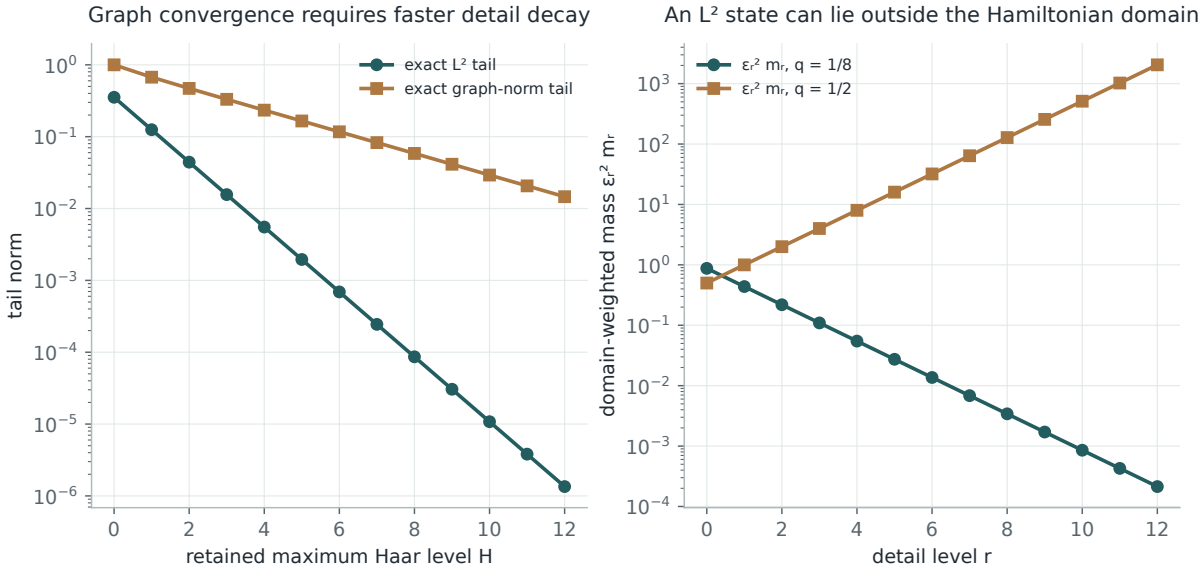


Figure 6. Exact infinite tails for the geometric level masses of Example 13.3. The operator domain controls a weighted tail in addition to the state norm tail. The parameters are declared in the plate and its reproducible Python source.

13.5 Compact resolvents and thermal weights

The omitted embedded resolvent has the exact norm

$$\|(I - \mathsf{E}_M)(z - H_\epsilon)^{-1}\| = \sup_{m > M} |z - \epsilon_m|^{-1}. \tag{119}$$

The upper bound follows by Parseval. For the lower bound, test on a unit vector at any omitted level and take the supremum. Since every level is nonzero and finite dimensional, this proves the following equivalence.

Proposition 13.4 (Exact compactness criterion). *The operator H_ϵ has compact resolvent if and only if $|\epsilon_m| \to \infty$. In that case the embedded finite resolvents converge in operator norm. This is an assertion about the embedded resolvents, rather than the zero-Hamiltonian extensions of* (116).

Proof. If the absolute energies tend to infinity, the right side of (119) tends to zero. The retained part of the resolvent has finite rank, so the resolvent is a norm limit of finite-rank operators and is compact. Conversely, if the absolute energies do not tend to infinity, there are infinitely many distinct levels m_j with $|\epsilon_{m_j}| \le C$ for some C. Unit vectors u_j on these levels are orthogonal. For a fixed nonreal z, their resolvent images are also orthogonal and have norm at least $(|z| + C)^{-1}$. They have no convergent subsequence. The resolvent is therefore not compact. □

When $|\epsilon_m| \to \infty$, the omitted part of the resolvent for $\widetilde{H}_M$ in (116) still carries the constant multiplier z^{-1}. It does not converge in norm to the omitted resolvent of H_ϵ. Strong resolvent convergence and norm convergence of an embedded finite-rank resolvent are distinct claims.

For $\beta > 0$, the multiplier $e^{-\beta H_\epsilon}$ is bounded precisely when the energies are bounded below. If there is no lower bound, the levelwise exponential is an unbounded operator with domain $\{f : \sum_m e^{-2\beta\epsilon_m} \|\mathsf{F}_m f\|_2^2 < \infty\}$. Thus a finite Euclidean intertwining identity does not by itself supply a bounded infinite heat semigroup. When bounded below, the positive operator has trace

$$\operatorname{Tr}(e^{-\beta H_\epsilon}) = e^{-\beta\epsilon_0} + \sum_{m\ge 1} (p-1)p^{m-1} e^{-\beta\epsilon_m}, \tag{120}$$

possibly infinite. This follows by summing its nonnegative diagonal entries in the complete Haar basis. It is trace class exactly when this series converges. For $\epsilon_m = cm$ with $c > 0$, the condition is $\beta c > \log p$, by the geometric-series test. For $\epsilon_m = p^{\alpha m}$ with $\alpha > 0$, it converges for every $\beta > 0$, since the exponential energy term eventually dominates the linear-in-m logarithm of the multiplicity. A thermodynamic partition function therefore needs an energy choice and a summability condition in addition to Hilbert-space reconstruction.

14 Compatible interactions across finitely many primes

14.1 Tensor embeddings and rectangular cutoffs

Let $\mathcal{P}$ be a finite prime set and let $\mathcal{K}_{\text{tor}}$ be a finite-dimensional complex space for the retained torsion and dyadic sign registers. Set

$$\mathcal{H}_{\mathcal{P}}^{\infty} = \mathcal{K}_{\text{tor}} \otimes \bigotimes_{p\in\mathcal{P}} L^2(\mathbb{Z}_p, \mu_p). \tag{121}$$

The corresponding finite register embeds by $I_{\mathcal{K}_{\text{tor}}} \otimes \bigotimes_p \mathcal{U}_{p,h_p}$. For a multi-depth $\mathbf{h} = (h_p)$, define

$$\mathsf{E}_{\mathbf{h}} = I_{\mathcal{K}_{\text{tor}}} \otimes \bigotimes_p \mathsf{E}_{p,h_p}.$$

Factorwise projection identities give

$$\mathsf{E}_{\mathbf{h}}\mathsf{E}_{\mathbf{k}} = \mathsf{E}_{\min(\mathbf{h},\mathbf{k})}, \qquad \mathsf{E}_{\mathbf{h}} f \to f \quad \text{as every } h_p \to \infty. \tag{122}$$

To justify the second statement, first take an elementary tensor. The difference of the tensor projections and the identity is a finite telescoping sum, one factor of which tends strongly to zero in each term,

while all other projection factors have norm at most one. Finite sums of elementary tensors are dense, and the projections are contractions. This extends convergence to every f.

Refinement in distinct coordinates is path independent because

$$(V_p \otimes I)(I \otimes V_q) = (I \otimes V_q)(V_p \otimes I) = V_p \otimes V_q.$$

The identity on $\mathcal{K}_{\mathrm{tor}}$ leaves the separately retained sign channel untouched by the averaging. This is a finite set of primes. An infinite tensor product, a restricted adelic product, and a model including the Archimedean place each require further choices.

Define the joint detail projections

$$\mathsf{F}_{\mathbf{r}} = I_{\mathcal{K}_{\mathrm{tor}}} \otimes \bigotimes_{p \in \mathcal{P}} \mathsf{F}_{p,r_p}, \qquad \mathbf{r} \in \mathbb{N}^{\mathcal{P}}. \tag{123}$$

They are pairwise orthogonal and have finite-dimensional ranges. Rectangular sums equal $\mathsf{E}_{\mathbf{h}}$, so their strong sum is the identity. This permits interactions between primes while preserving the exact depth filtration.

14.2 A complete block-domain theorem

The scalar weights are one particularly simple choice. The following editorial extension states the exact additional input for interacting blocks. It is useful because the torsion operators inside a joint packet need not commute with one another.

Theorem 14.1 (Self-adjoint packet blocks and compatible finite dynamics)**.** *For every* $\mathbf{r}$*, let* $A_{\mathbf{r}}$ *be a self-adjoint operator on the finite-dimensional space* $\mathcal{K}_{\mathbf{r}} = \operatorname{Ran} \mathsf{F}_{\mathbf{r}}$*. On* $\mathcal{H}_{\mathcal{P}}^{\infty} = \bigoplus_{\mathbf{r}} \mathcal{K}_{\mathbf{r}}$ *define*

$$\begin{aligned} \mathcal{D}(H_A) &= \left\{ f : \sum_{\mathbf{r}} \left\| A_{\mathbf{r}} \mathsf{F}_{\mathbf{r}} f \right\|^2 < \infty \right\}, \\ H_A f &= \sum_{\mathbf{r}} A_{\mathbf{r}} \mathsf{F}_{\mathbf{r}} f. \end{aligned} \tag{124}$$

Then H_A *is self-adjoint, and finite rectangular packet sums form a core. Every rectangular range reduces* H_A*, and its finite restriction* $H_A^{(\mathbf{h})}$ *satisfies exact embedding intertwining with every larger rectangular restriction. For bounded Borel* F*,*

$$F(H_A) f = \sum_{\mathbf{r}} F(A_{\mathbf{r}}) \mathsf{F}_{\mathbf{r}} f, \qquad F(H_A)\mathsf{E}_{\mathbf{h}} \to F(H_A) \quad \textit{strongly}. \tag{125}$$

In particular,

$$\left\| e^{-itH_A} (I - \mathsf{E}_{\mathbf{h}}) f \right\| = \left\| (I - \mathsf{E}_{\mathbf{h}}) f \right\|. \tag{126}$$

Proof. Finite rectangular sums belong to the domain and are dense. Symmetry holds by symmetry on every block. If g belongs to the adjoint domain and $k = H_A^* g$, testing against an arbitrary vector in $\mathcal{K}_{\mathbf{r}}$ gives $\mathsf{F}_{\mathbf{r}} k = A_{\mathbf{r}} \mathsf{F}_{\mathbf{r}} g$. Square summability of k is precisely the domain condition in (124). Thus $H_A^* = H_A$. Truncating the two square-summable sequences $f_{\mathbf{r}}$ and $A_{\mathbf{r}} f_{\mathbf{r}}$ proves the graph-core assertion.

The rectangular projection selects a collection of whole blocks, so it commutes with H_A on the domain and reduces the operator. The restriction to each retained block is literally the same matrix at every subsequent depth. This proves the finite intertwining, and finite-dimensional spectral calculus

on each block gives (125). Boundedness of F bounds its omitted image by $\|F\|_\infty \|(I - \mathsf{E}_{\mathbf{h}})f\|$, proving strong convergence. Finally, each block exponential is unitary, so orthogonality of the omitted blocks gives (126). □

For example, on a joint depth packet one may take

$$A_{\mathbf{r}} = \left(\sum_a \epsilon_{a,\mathbf{r}} T_a \right) \otimes I_{\text{phase},\mathbf{r}}, \qquad \epsilon_{a,\mathbf{r}} \in \mathbb{R}, \quad T_a = T_a^*.$$

The matrices T_a act on the sign and torsion factors. Their mutual commutation is unnecessary because their sum is self-adjoint on the finite block. The coefficients assigned to an existing packet are fixed under refinement. An extension acting wholly inside a new detail complement has no matrix element between that complement and the old range. A cross term between them would be a different model.

There is also an exact construction for any *specified* self-adjoint compatible finite family along a cofinal nested sequence. Suppose finite-dimensional spaces $\mathcal{V}_0 \subset \mathcal{V}_1 \subset \cdots$ have dense union, and self-adjoint K_M satisfy $K_{M+1} f = K_M f$ for $f \in \mathcal{V}_M$. Self-adjointness implies that $\mathcal{V}_M^\perp \cap \mathcal{V}_{M+1}$ is invariant: for g there and $f \in \mathcal{V}_M$, $\langle f, K_{M+1} g\rangle = \langle K_{M+1} f, g\rangle = 0$. The initial space $\mathcal{V}_0$ and the successive difference spaces therefore have stable self-adjoint finite blocks. Their orthogonal direct sum with the maximal square-summability domain is self-adjoint by the same adjoint proof above. This is an editorial compatible-family construction, not an assertion that arbitrary finite Hamiltonians are compatible. The identities $K_{M+1}|_{\mathcal{V}_M} = K_M$ are its substantive hypothesis.

14.3 An exact cross-prime entangling generator

Let Z_{sign} have eigenvalues $+1, -1$ on a dyadic sign qubit, and let $B = B^*$ act on the remaining factors. Assume there are orthonormal states $|b_0\rangle, |b_1\rangle$ with $B|b_0\rangle = 0$ and $B|b_1\rangle = |b_1\rangle$. One concrete choice is an odd-prime torsion involution times a fixed odd-prime detail projector, with the torsion state in its $+1$ eigenspace. For real $g \neq 0$, use

$$H_{\text{int}} = g Z_{\text{sign}} \otimes B, \qquad |\Psi(0)\rangle = |+_x\rangle \otimes \frac{|b_0\rangle + |b_1\rangle}{\sqrt{2}}.$$

Evolution under this explicit interaction alone gives

$$|\Psi(t)\rangle = \frac{1}{\sqrt{2}} \left(|+_x\rangle \otimes |b_0\rangle + e^{-igtZ_{\text{sign}}} |+_x\rangle \otimes |b_1\rangle \right). \tag{127}$$

Tracing out the orthogonal b states yields the equally weighted mixture of the two displayed sign states. Their overlap is $\cos(gt)$. A mixture of two unit vectors with equal weights and overlap magnitude c has eigenvalues $(1 + c)/2$ and $(1 - c)/2$: its trace is one and its determinant on their two-dimensional span is $(1 - c^2)/4$. Thus the sign entropy in bits is

$$S_{\text{sign}}(t) = h_2 \left(\frac{1 + |\cos(gt)|}{2} \right), \quad h_2(u) = -u \log_2 u - (1 - u) \log_2 (1 - u), \tag{128}$$

with $0 \log_2 0 = 0$. At $t_* = \pi/(2|g|)$ the two sign states are orthogonal and the entropy is one bit. A fixed level projector is compatible under refinement, so this interaction can entangle the prime factors while preserving the cylinder embedding. The tensor factorization alone creates no entanglement. It is this chosen generator and initial superposition that produce the result.

15 Controlled leakage when compatibility is broken

The preceding identities are exact because the coarse range reduces the Hamiltonian. A weighted coupling to its complement has a different, quantifiable behavior. This section concerns finite-dimensional self-adjoint operators; no unbounded perturbation theorem is implicit.

Let V be an isometry from a coarse space to a finite fine space, set $P = VV^*$ and $Q = I - P$, and let $H_0 = H_0^*$ satisfy $H_0V = VH_c$. Let $B = B^*$ be purely off diagonal:

$$PBP = 0, \qquad QBQ = 0.$$

For a real λ, define $H_\lambda = H_0 + \lambda B$. In the source model $\lambda = p^{-w}$ for an assigned weight $w \geq 1$.

Proposition 15.1 (Defect, leakage, and a second-order self-energy)**.** *Under these hypotheses,*

$$\|H_\lambda V - VH_c\| = |\lambda| \, \|BV\| , \tag{129}$$

$$\left\|Qe^{-itH_\lambda}P\right\| \leq \min\{1, |t\lambda| \, \|QBP\|\}. \tag{130}$$

For $z \notin \sigma(QH_\lambda Q|_{Q\mathcal{H}})$, the coarse self-energy is

$$\Sigma_P(z) = PH_\lambda Q(z - QH_\lambda Q|_{Q\mathcal{H}})^{-1} QH_\lambda P, \tag{131}$$

where the inverse acts only on $Q\mathcal{H}$, and

$$\|\Sigma_P(z)\| \leq \frac{|\lambda|^2 \, \|PBQ\|^2}{\operatorname{dist}(z, \sigma(QH_\lambda Q|_{Q\mathcal{H}}))}. \tag{132}$$

Proof. The defect is λBV, so the first identity is immediate. Self-adjointness of H_0 makes its invariant coarse range reducing. Hence $Qe^{-itH_0}P = 0$, and Duhamel's identity gives

$$Qe^{-itH_\lambda}P = -i\lambda \int_0^t Qe^{-i(t-s)H_\lambda} Be^{-isH_0} P \, ds.$$

The off-diagonal assumption and reduction by H_0 give $Be^{-isH_0}P = QBPe^{-isH_0}P$. The two propagators and the exterior projections are contractions. Integrating the norm bound over an interval of length $|t|$ gives the second expression in (130). The bound one follows directly from unitarity and the norms of the projections.

The coarse/detail blocks of H_0 vanish, so $PH_\lambda Q = \lambda PBQ$ and $QH_\lambda P = \lambda QBP$. The detail compression is self-adjoint. Diagonalizing it bounds the resolvent norm by the reciprocal spectral distance. Since $QBP = (PBQ)^*$, submultiplicativity gives (132). □

The small-time leading term is $-it\lambda QBP$. More precisely,

$$\left\|Qe^{-itH_\lambda}P + it\lambda QBP\right\| \leq \frac{|\lambda|t^2}{2}(\|H_\lambda\| + \|H_0\|) \, \|QBP\| . \tag{133}$$

To verify this, subtract the constant integrand QBP in Duhamel's formula and use $\left\|e^{-iuK} - I\right\| \leq |u| \, \|K\|$ for a finite self-adjoint K. The two deviations are bounded by $(|t - s| \, \|H_\lambda\| + |s| \, \|H_0\|) \, \|QBP\|$ on the oriented integration interval. Their integrals are both $t^2/2$. For $|\lambda| \leq 1$, this is a uniform $O(t^2|\lambda|)$

remainder. A generic $O(t^2\lambda^2)$ remainder would omit the first-order interaction with the background Hamiltonian.

For the stipulated assignment $\lambda = p^{-w}$, the operator defect and this leakage upper bound scale linearly with p^{-w}. The self-energy bound is quadratic in p^{-w} away from the detail spectrum. Under the purely off-diagonal hypothesis, $QH_\lambda Q = QH_0Q$, so the denominator does not depend on λ. Approaching a detail resonance removes that uniform estimate. These are exact finite quantum statements for a chosen coupling law. An integral valuation packet does not, by itself, force every physical interaction strength to equal p^{-w}.

16 A local walk on physical position and refinement depth

16.1 The finite graph and its unitary tick

Let $N \geq 3$ and $h \geq 0$. Let C_N be the physical cycle and let $T_{p,h}$ be the rooted finite p-ary tree of prefixes of length at most h. The product graph is

$$\mathcal{G} = C_N \square T_{p,h}.$$

A vertex is $v = (x, \mathbf{d})$, with $x \in \mathbb{Z}/N\mathbb{Z}$ and $\mathbf{d}$ a prefix. An edge changes either x by one on the cycle or the prefix by one parent-child step. The graph has $N \sum_{r=0}^{h} p^r$ vertices. It includes all tree depths, unlike a cylinder register which contains only the residues at its last depth.

Let $\vec{E}$ denote the directed arcs and let $\tau_2 = 2$, $\tau_p = p - 1$ for odd p. The finite arc Hilbert space is

$$\mathcal{H}_{\text{arc}} = \ell^2(\vec{E}) \otimes \mathbb{C}^{\tau_p}.$$

For each vertex of degree d_v, define the normalized outgoing vector $s_v = d_v^{-1/2}(1, \dots, 1)$ and the Grover coin $G_v = 2|s_v\rangle\langle s_v| - I$. Its square is I, because $|s_v\rangle\langle s_v|$ is a projection. Take any local unitary R_v on the torsion or sign factor and put

$$C = \bigoplus_v (G_v \otimes R_v).$$

The reversal shift S sends the arc $v \to u$ to $u \to v$, retaining the internal register. It is a permutation with $S^* = S$ and $S^2 = I$. The one-tick walk

$$U_{p,h,N} = SC \tag{134}$$

is unitary because $U_{p,h,N}^* U_{p,h,N} = C^* S^* S C = C^* C = I$. Here locality concerns the graph edges, with no continuum limit assumed.

16.2 Local conservation and the product cone

Write $\psi^n(v,u)$ for the internal amplitude on an arc leaving v before the coin, and $\phi^n = C\psi^n$ for its post-coin value. Define

$$\rho_v^n = \sum_{u \sim v} \|\psi^n(v,u)\|^2, \qquad F_{v\to u}^n = \|\phi^n(v,u)\|^2.$$

The local unitary coin gives $\sum_u F^n_{v\to u} = \rho^n_v$. The reversal shift gives $\rho^{n+1}_v = \sum_u F^n_{u\to v}$. Subtraction proves

$$\rho^{n+1}_v - \rho^n_v = \sum_{u\sim v}(F^n_{u\to v} - F^n_{v\to u}). \tag{135}$$

For every set of vertices A, summing this equation cancels every internal directed flux once with its opposite occurrence. Hence

$$\sum_{v\in A}(\rho^{n+1}_v - \rho^n_v) = \sum_{\substack{u\notin A,\ v\in A\\ u\sim v}} F^n_{u\to v} - \sum_{\substack{v\in A,\ u\notin A\\ u\sim v}} F^n_{v\to u}. \tag{136}$$

Taking A to be all vertices gives conservation of total probability. The coarse identity uses a subset of vertices and does not presume amplitude-level Haar refinement.

Proposition 16.1 (Exact finite propagation)**.** *If the initial outgoing amplitudes are supported at v_0, then*

$$d_{C_N}(x, x_0) + d_T(\mathbf{d}, \mathbf{d}_0) > n \quad\Longrightarrow\quad \rho^n_{(x,\mathbf{d})} = 0. \tag{137}$$

For two prefixes, writing lcp *for the length of their longest common prefix,*

$$d_T(\mathbf{d}, \mathbf{d}_0) = |\mathbf{d}| + |\mathbf{d}_0| - 2\,\mathrm{lcp}(\mathbf{d}, \mathbf{d}_0).$$

In particular, each tick allows at most one physical edge or one tree edge, and $\big||\mathbf{d}| - |\mathbf{d}_0|\big| \le n$.

Proof. Every path in a Cartesian product graph uses a number of physical edges at least the physical distance and a number of tree edges at least the tree distance. Their sum is a lower bound on path length. Following a shortest physical path and then a shortest tree path attains that sum, so it is the product graph distance. In a rooted tree the shortest path rises to the longest common prefix and then descends, giving the displayed formula.

The claim at $n = 0$ is the support hypothesis. A coin mixes outgoing arcs only at their common source, so it does not enlarge the vertex support. The shift moves the source across one edge. A state supported in the radius-n ball is therefore supported in the radius-$(n+1)$ ball after a tick. Induction proves the result. Since each graph edge changes only one factor, the individual distance and depth bounds follow. □

The depth difference alone is weaker than the full tree distance. Different prefixes at the same depth can have positive distance, so a plot aggregated over depth does not show the complete graph geometry. The cone is an exact support property of this finite coined model. A Lorentzian continuum dispersion law or a relativistic limiting field equation requires a separate scaling analysis.

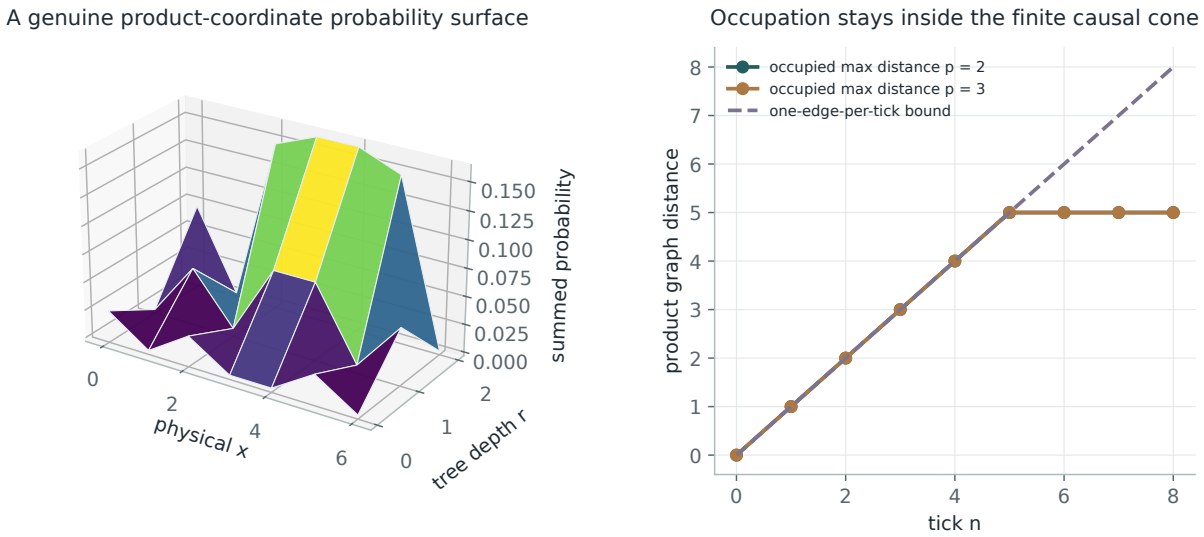


Figure 7. A finite cycle–tree walk, aggregated over vertices at the same tree depth. The graph-distance support check uses the full product graph, including prefix labels. Probability plotted against depth alone displays only an aggregation of that geometry.

16.3 Why growing this tree is not automatically Haar refinement

Example 16.2 (A boundary coin breaks the canonical arc embedding)**.** Take $h \geq 1$ and $N \geq 3$, and embed the old arcs of $C_N \square T_{p,h}$ in $C_N \square T_{p,h+1}$ by identical labels, with zero amplitude on new arcs. Denote this isometry by J. At an old leaf, the old degree is $d = 3$: two cycle neighbors and one parent. The larger graph adds p children, changing the degree to $d + p$. For an old outgoing arc basis vector e_j, the new Grover coin gives amplitude $2/(d + p)$ on each of the p new outgoing child arcs. An internal unitary preserves its internal norm. The subsequent reversal shift sends these amplitudes to new arcs at the children, outside the old arc range. Therefore, for a unit internal state χ,

$$\|(I - JJ^*)U_{p,h+1,N}J(e_j \otimes \chi)\| = \frac{2\sqrt{p}}{3+p} > 0. \tag{138}$$

The state $JU_{p,h,N}(e_j \otimes \chi)$ has zero amplitude there, so $U_{p,h+1,N}J \neq JU_{p,h,N}$.

Each walk remains unitary, locally conservative, and confined to its product graph cone. A different coin family, embedding, or infinite-tree construction could be investigated. The canonical old-arc inclusion fails the compatibility hypothesis of Theorem 14.1. The cylinder limit and the tree walk are exact models with distinct state spaces.

17 The boundary of the infinite-depth result

The exact result comprises complex L^2 cylinder reconstruction and dynamics for declared real level energies or self-adjoint packet blocks. State convergence uses an L^2 tail; graph norm convergence uses an operator-domain tail. Compact resolvents require energies escaping every bounded set, and finite thermal traces require multiplicity summability. Each claim has a distinct hypothesis.

The source programs check finite restrictions of these formulas. Code 45 uses three towers,

$$(p, M) = (2, 7),\ (3, 5),\ (5, 3),$$

and a fixed random seed. Their terminal zero tails do not certify an infinite domain or compactness; the proofs supply those conclusions. Code 40 tests specified dyadic/ternary interactions and weighted off-diagonal couplings. Code 42 tests finite product walks, whose unitarity does not imply compatibility with the Haar tower.

C15 records finite refinement identities and C16 the conditional infinite-depth construction. Their archived chapter numbers predate the current arrangement. Exact members are chapter 31 for multi-prime dynamics, chapter 33 for product walks, and chapter 36 for the infinite Haar limit. Prime-polyphase semigroups appear in the spectral part. No fresh Lean compilation, unrestricted quantum-field completion, infinite-prime adelic construction, or experimental identification is claimed here.

18 What refinement establishes, and what remains

The finite construction establishes a concrete chain of statements. Unit coordinates factor a finite arithmetic register. Normalized Haar averages identify a coarse Hilbert space inside a fine one. A compatible Hermitian Hamiltonian then carries the same evolution on that range. This last step uses compatibility, rather than normalization alone. When a new coupling breaks the reducing-range property, the hidden sector produces leakage and a nontrivial self-energy.

The spectral construction makes that distinction calculable. Polyphase coordinates display the fine operator's channels without removing any of them. Radial or residue elimination introduces an energy-dependent effective operator on a regular resolvent chart. The memory formulation retains the hidden initial state and the causal history. Chebyshev decimation describes an exact spectral relation where its Dirichlet inverse exists. Hypergeometric companion functions organize entire power-series identities in finite or bounded settings, with additional domains when unbounded operators enter.

The infinite construction is a theorem for a declared hierarchical model. Probability Haar measure and normalized cylinders give a common Hilbert space. Orthogonal depth blocks and self-adjoint weights specify its maximal Hamiltonian domain. Finite compatible blocks then approximate the state evolution strongly. Graph-norm convergence concerns vectors in that domain. Operator-norm convergence and compact resolvent require stronger spectral conditions. A finite walk whose old vertices change degree under extension is outside this compatibility theorem until a suitable comparison map is supplied.

> **The remaining research interfaces**
>
> The analytic adic hierarchy still needs an exact formal translation if it is to carry a Lean certificate. General depth-dependent walks need their own limit argument. A physical fiber theory needs a covariant coupling, a state-selection or production rule, and consistent energy accounting. Each interface is an additional claim with its own hypotheses.

The visual atlas and verification companion make the scope reviewable. Their finite examples test identities, illuminate domain distinctions, and retain negative controls. They do not extrapolate a bounded calculation into an infinite proof. Related archived formal results establish normalized cells, exact tower properties, and dyadic reconstruction statements. They are useful mathematical interfaces, but their theorem names and hypotheses do not certify every result in this volume.

A candidate fiber interpretation must specify how refinement degrees of freedom couple to spacetime and how they are populated. The covariant fiber chapter addresses that question before the cosmological calculations are used. Its energy-budget obstruction remains controlling: downstream transport from a prescribed distribution is meaningful, but it is not evidence that the proposed production history generated that distribution.

Sources and primary context

Project sources

[S1] Komorebi P. *Projector-Refinement Physics, Volume I*. The arithmetic source studies cover unit registers, Haar refinement, finite-adelic dynamics, product walks, radial self-energy, polyphase structure, infinite Haar limits, spectral renormalization, and hypergeometric companion functions.

[S2] Komorebi P. *Volume I Technical Supplement and Independent Review Dossier*. The arithmetic claim dossiers and formula registries specify the scope of the corresponding derivations and computational checks.

[S3] Komorebi P. *Volume I Second Implementation Reproduction*. Finite Haar tests provide internal reproduction evidence for their stated inputs, not for the entire infinite theory.

[S4] Komorebi P. *Komorebi Physics formal source*. The verification appendix identifies related tower and concrete reconstruction results. Shared interfaces are not a certificate for every arithmetic theorem.

Primary context

[C1] NIST Digital Library of Mathematical Functions, §16.2 and §18.5. Generalized hypergeometric series and Chebyshev conventions. Relevant definitions and convergence hypotheses inspected.

[C2] G. Teschl, *Jacobi Operators and Completely Integrable Nonlinear Lattices*, Mathematical Surveys and Monographs 72, American Mathematical Society, 2000. Author's online edition, §§1.1 and 2.1. Jacobi continued-fraction and Weyl-function context. The sign of a Weyl function changes when $(J - z)^{-1}$ is replaced by $(z - J)^{-1}$.

[C3] S. V. Kozyrev, "Wavelet theory as p-adic spectral analysis," *Izvestiya: Mathematics* 66(2), 367–376 (2002). doi:10.1070/IM2002v066n02ABEH000381. Published English Theorems 2 and 7 inspected. The $\mathbb{Q}_p$ Vladimirov setting supplies wavelet context; the present $\mathbb{Z}_p$ level-diagonal construction is proved separately.

[C4] D. Aharonov, A. Ambainis, J. Kempe and U. Vazirani, "Quantum Walks on Graphs," *Proceedings of STOC 2001*, 50–59. doi:10.1145/380752.380758. arXiv:quant-ph/0012090v2, 25 May 2002, §3.1 inspected. Standard coined-walk context; the product-graph arc construction and its continuity identities are proved here.

[C5] P. P. Vaidyanathan, "Multirate digital filters, filter banks, polyphase networks, and applications: a tutorial," *Proceedings of the IEEE* 78(1), 56–93 (1990). doi:10.1109/5.52200. Caltech author-repository record. Metadata and abstract inspected; full PDF retrieval failed. Polyphase history and terminology only.

Scope of the literature check

The hypergeometric bottom parameters used here are positive, so the nonpositive-integer exclusions in §16.2 do not arise. The Chebyshev formulas use the standard first- and second-kind definitions. These scalar conventions do not supply an unbounded-operator domain.

The cited Teschl, Kozyrev, and Aharonov–Ambainis–Kempe–Vazirani passages are used only for the contextual comparisons stated above. Their settings differ from the finite tree, compact adic hierarchy, and irregular arc graph of this volume. The comparison is limited to the explicit uses recorded for each reference. The Vaidyanathan PDF could not be retrieved; its metadata and author abstract establish only the stated historical context. All exact model propositions in this volume carry internal derivations. No exhaustive novelty search or literature-priority claim is made.